McGann

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Summary

Methodology

Significant Findings

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.1.6
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.1     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.2.0
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom

Attaching package: 'zoo'


The following objects are masked from 'package:base':

    as.Date, as.Date.numeric
```

Full Analysis / Results

Below are several tables outlining each of the categories of analysis requested.

Question1	Question2	P_value	Significant	Test	Adj_P	Sig_Adj
Q8h	role	5e-04	TRUE	fisher	0.0080	TRUE
Q9	role	0.0015	TRUE	fisher	0.0120	TRUE
Q8a	role	0.003	TRUE	fisher	0.0160	TRUE
Q8k	role	0.0075	TRUE	fisher	0.0300	TRUE
Q8e	role	0.0295	TRUE	fisher	0.0731	FALSE
Q8g	role	0.032	TRUE	fisher	0.0731	FALSE
Q11a	role	0.0255	TRUE	fisher	0.0731	FALSE
Q8d	role	0.043	TRUE	fisher	0.0860	FALSE
Q8c	role	0.0505	FALSE	fisher	0.0898	FALSE
Q8j	role	0.0735	FALSE	fisher	0.1176	FALSE
Q8b	role	0.1019	FALSE	fisher	0.1482	FALSE
Q11b	role	0.1294	FALSE	fisher	0.1725	FALSE
Q10a	role	0.1514	FALSE	fisher	0.1863	FALSE
Q8f	role	0.1654	FALSE	fisher	0.1890	FALSE
Q10b	role	0.1864	FALSE	fisher	0.1988	FALSE
Q8i	role	0.4023	FALSE	fisher	0.4023	FALSE

Question1	Question2	P_value	Significant	Test	Adj_P	Sig_Adj
Q14	role	0.0515	FALSE	fisher	0.1545	FALSE
Q13	role	0.1849	FALSE	fisher	0.2774	FALSE
Q12	role	0.3333	FALSE	fisher	0.3333	FALSE

Resident/Dentist Comfort & Clinical Practice Analysis

Below is a table comparing residents and dentists comfort levels and clinical practices.

Resident/Dentist Knowledge Analysis

Below is a table comparing residents and dentist questions on knowledge.

Residents/Dentists Quiz Analysis

Below is a table comparing residents and dentist quiz results.

Environment Analysis

Below is a table comparing environmental questions.

Question1	Question2	P_value	Significant	Test	Adj_P	Sig_Adj
Q15	role	0.8851	FALSE	fisher	0.8851	FALSE
Q16	role	0.7776	FALSE	fisher	0.8851	FALSE
Q17	role	0.8301	FALSE	fisher	0.8851	FALSE
Q18	role	0.5447	FALSE	fisher	0.8851	FALSE
Q19	role	0.6367	FALSE	fisher	0.8851	FALSE
Q20	role	0.2224	FALSE	fisher	0.8851	FALSE
score	role	0.4813	FALSE	fisher	0.8851	FALSE

Question1	Question2	P_value	Significant	Test	Adj_P	Sig_Adj
Q8a	Q5	0.0195	TRUE	fisher	0.0918	FALSE
Q8g	Q5	0.0155	TRUE	fisher	0.0918	FALSE
Q8h	Q5	0.0235	TRUE	fisher	0.0918	FALSE
Q8k	Q5	0.027	TRUE	fisher	0.0918	FALSE
Q13	Q5	0.02	TRUE	fisher	0.0918	FALSE
Q8c	Q5	0.038	TRUE	fisher	0.1077	FALSE
Q9	Q5	0.1724	FALSE	fisher	0.4187	FALSE
Q8b	Q5	0.2409	FALSE	fisher	0.4954	FALSE
Q8f	Q5	0.2624	FALSE	fisher	0.4954	FALSE
Q10a	Q5	0.2914	FALSE	fisher	0.4954	FALSE
Q8d	Q5	0.5692	FALSE	fisher	0.8064	FALSE
Q8e	Q5	0.5392	FALSE	fisher	0.8064	FALSE
Q8i	Q5	0.7096	FALSE	fisher	0.8229	FALSE
Q8j	Q5	0.7261	FALSE	fisher	0.8229	FALSE
Q10b	Q5	0.6607	FALSE	fisher	0.8229	FALSE
Q12	Q5	0.9835	FALSE	fisher	0.9905	FALSE
score	Q5	0.9905	FALSE	fisher	0.9905	FALSE

Question1	Question2	P_value	Significant	Test	Adj_P	Sig_Adj
Q9	Q3	0.004	TRUE	fisher	0.0495	TRUE
score	Q3	0.0055	TRUE	fisher	0.0495	TRUE
Q8d	Q3	0.025	TRUE	fisher	0.1170	FALSE
Q10b	Q3	0.026	TRUE	fisher	0.1170	FALSE
Q8b	Q3	0.0365	TRUE	fisher	0.1314	FALSE
Q8a	Q3	0.0535	FALSE	fisher	0.1605	FALSE
Q8f	Q3	0.081	FALSE	fisher	0.1900	FALSE
Q8i	Q3	0.095	FALSE	fisher	0.1900	FALSE
Q12	Q3	0.0925	FALSE	fisher	0.1900	FALSE
Q10a	Q3	0.2619	FALSE	fisher	0.4714	FALSE
Q8j	Q3	0.3268	FALSE	fisher	0.5348	FALSE
Q8c	Q3	0.4528	FALSE	fisher	0.5536	FALSE
Q8e	Q3	0.4208	FALSE	fisher	0.5536	FALSE
Q8h	Q3	0.4283	FALSE	fisher	0.5536	FALSE
Q11a	Q3	0.4613	FALSE	fisher	0.5536	FALSE
Q11b	Q3	0.5657	FALSE	fisher	0.6364	FALSE
Q8g	Q3	0.6032	FALSE	fisher	0.6387	FALSE
Q8k	Q3	0.8821	FALSE	fisher	0.8821	FALSE

Dentist/Practice Analysis

Below is a table comparing dentists and their practices.

Additional Analysis

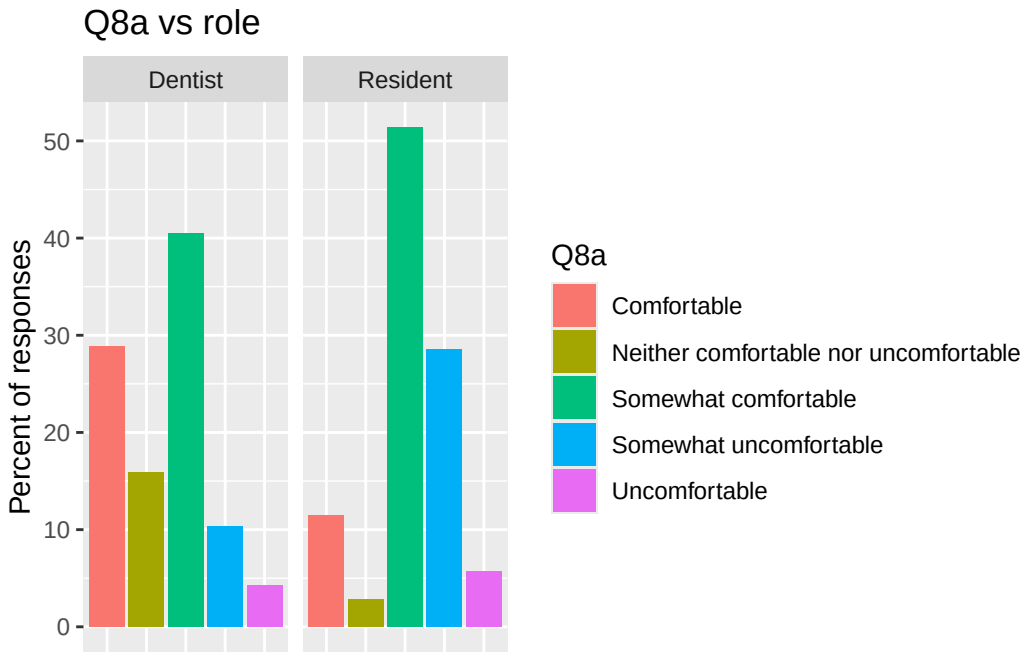
Below is a table comparing additional analysis questions.

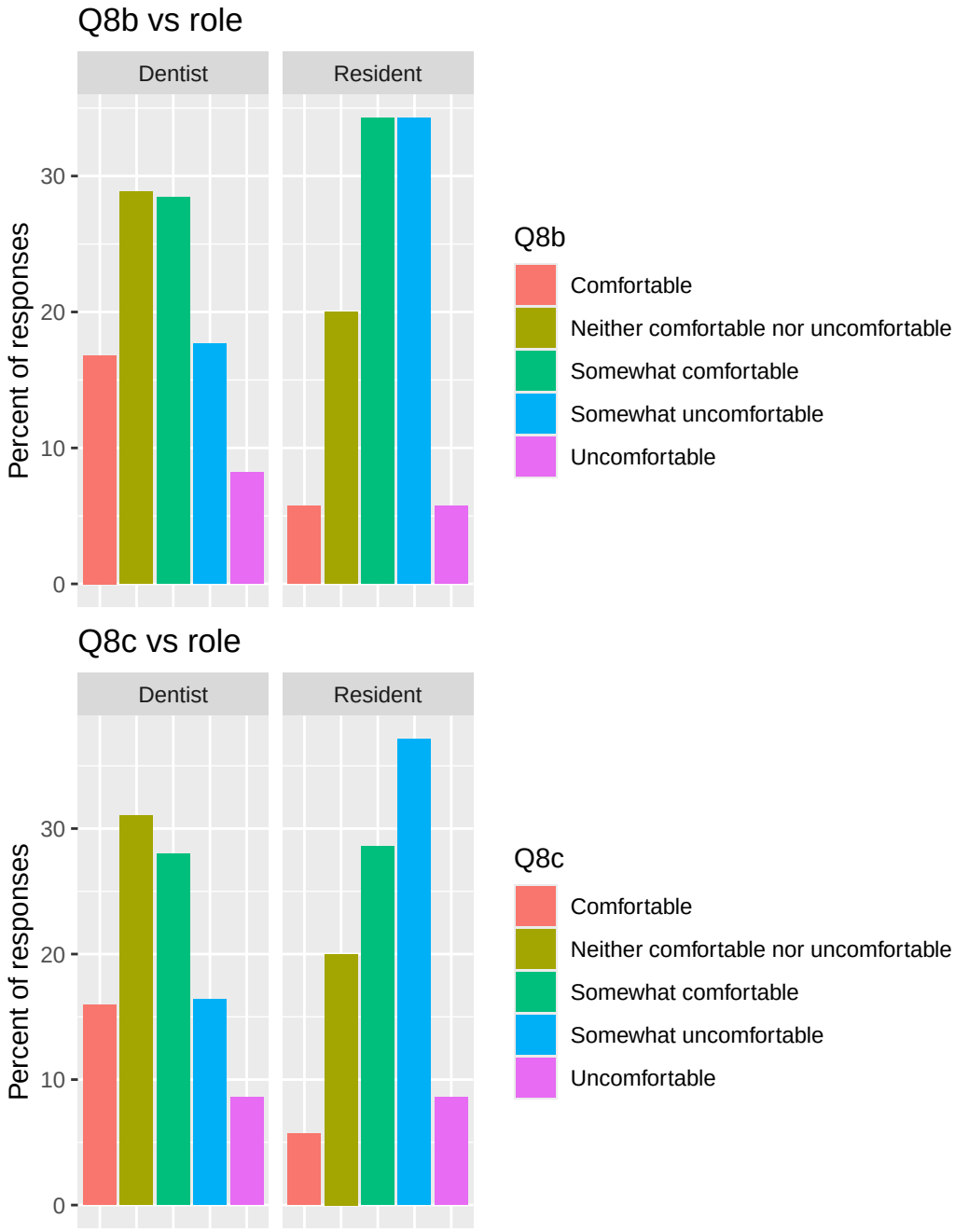
Appendix A: Plots

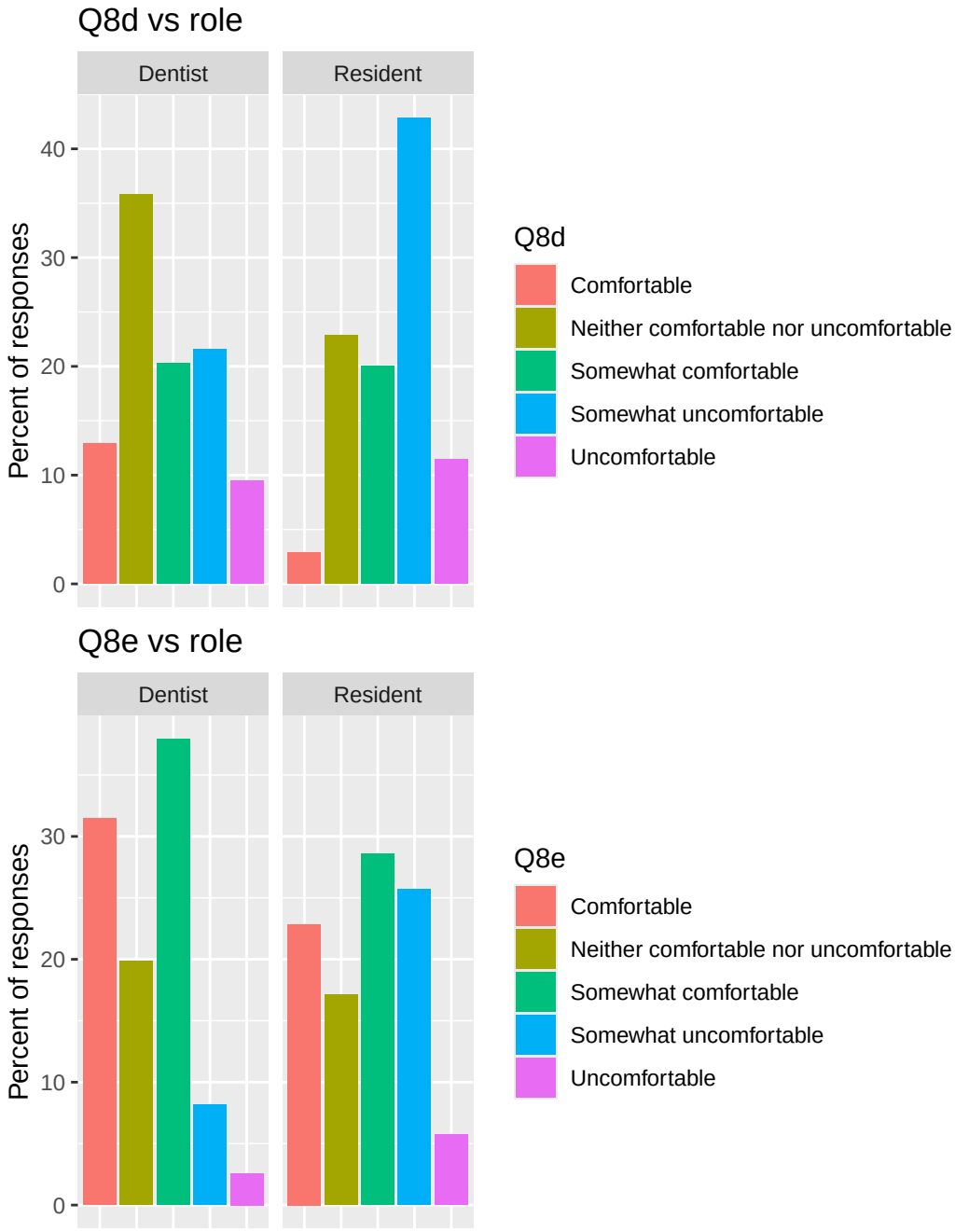
Comfort & Clinical Practice Plots

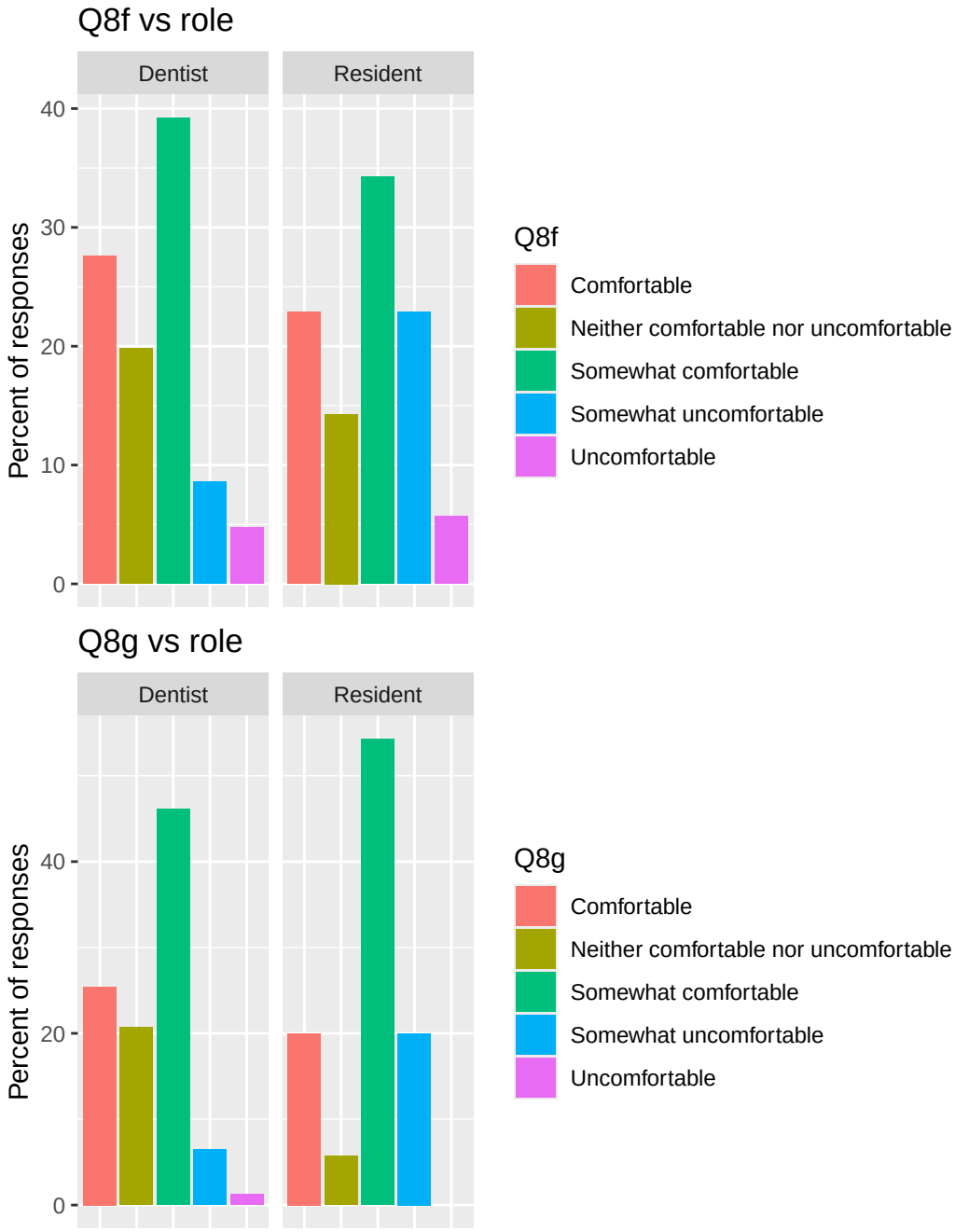
Below are plots comparing the comfort and clinical practices between residents and dentists.

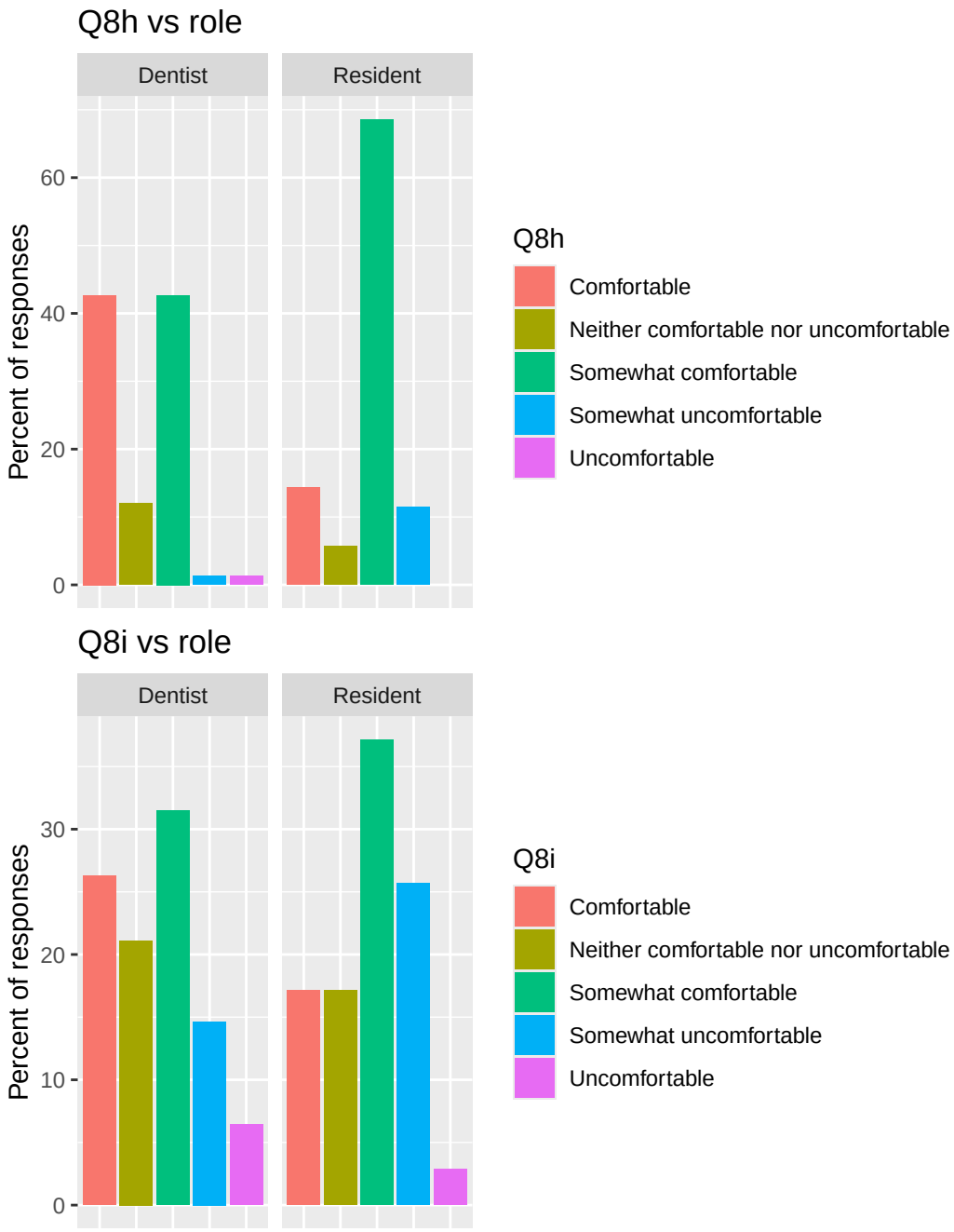
Question1	Question2	P_value	Significant	Test	Adj_P	Sig_Adj
Q8a	Q12	5e-04	TRUE	fisher	0.0005	TRUE
Q8b	Q12	5e-04	TRUE	fisher	0.0005	TRUE
Q8d	Q12	5e-04	TRUE	fisher	0.0005	TRUE
Q8e	Q12	5e-04	TRUE	fisher	0.0005	TRUE
Q8f	Q12	5e-04	TRUE	fisher	0.0005	TRUE
Q8g	Q12	5e-04	TRUE	fisher	0.0005	TRUE
Q8h	Q12	5e-04	TRUE	fisher	0.0005	TRUE
Q8i	Q12	5e-04	TRUE	fisher	0.0005	TRUE
Q8j	Q12	5e-04	TRUE	fisher	0.0005	TRUE
Q8k	Q12	5e-04	TRUE	fisher	0.0005	TRUE
Q9	Q12	5e-04	TRUE	fisher	0.0005	TRUE
Q10a	Q12	5e-04	TRUE	fisher	0.0005	TRUE
Q10b	Q12	5e-04	TRUE	fisher	0.0005	TRUE
Q11a	Q9	5e-04	TRUE	fisher	0.0005	TRUE
Q11b	Q9	5e-04	TRUE	fisher	0.0005	TRUE
Q10a	Q13	5e-04	TRUE	fisher	0.0005	TRUE
Q10b	Q13	5e-04	TRUE	fisher	0.0005	TRUE
Q8c	Q12	0.0015	TRUE	fisher	0.0015	TRUE

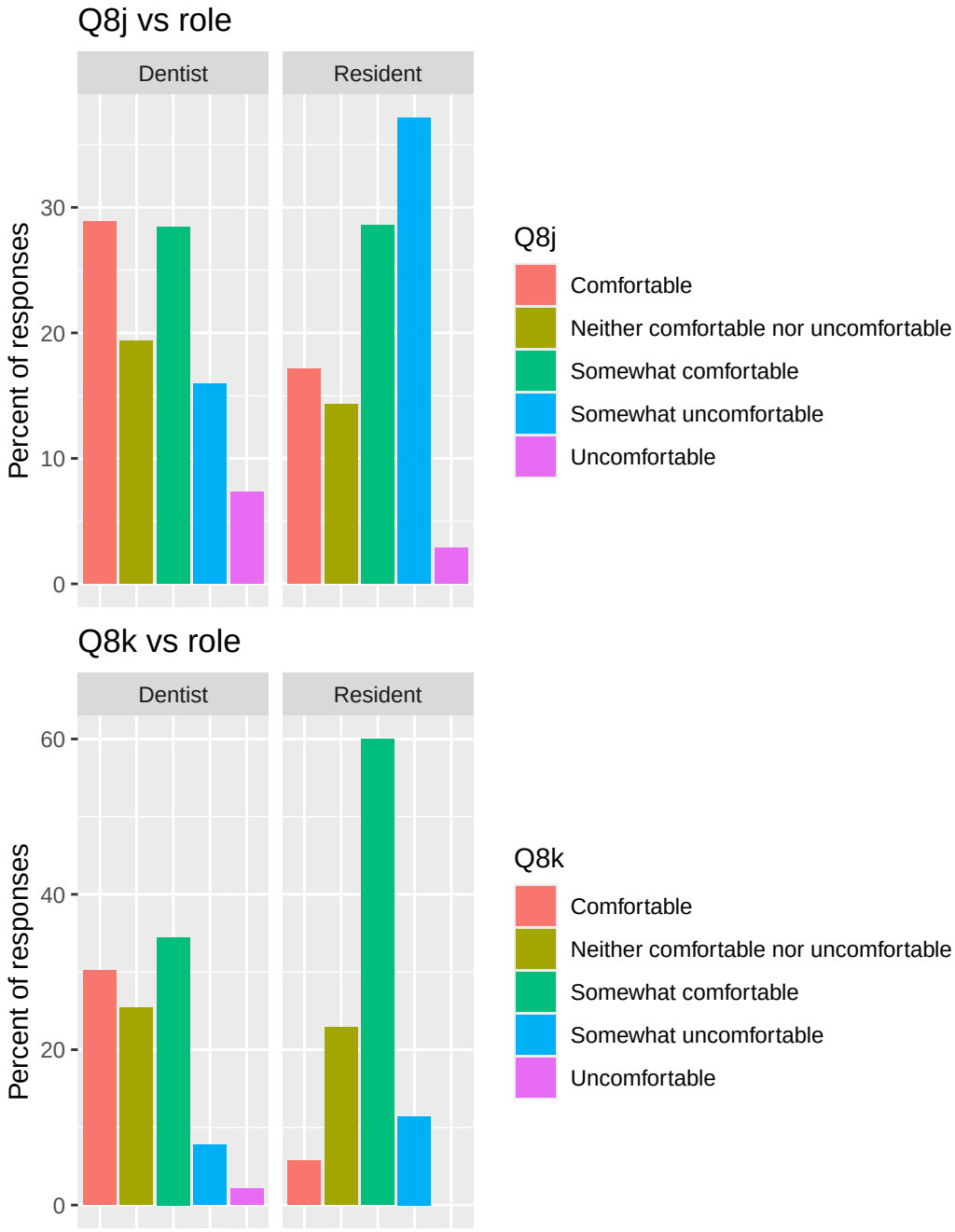


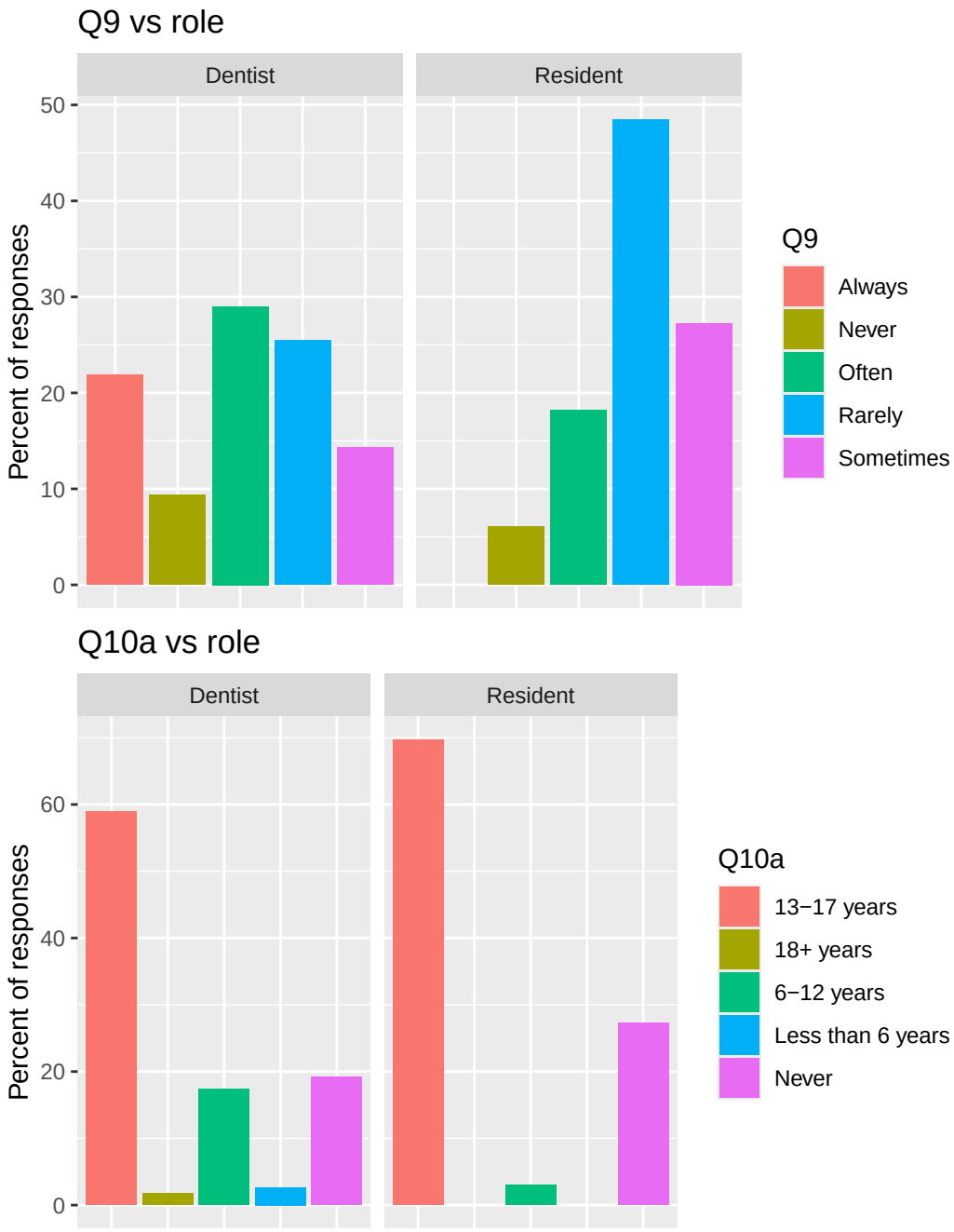

Q8b vs role
Dentist
Resident
Percent of responses
30
20
10
0
Q8b
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8c vs role
Dentist
Resident
Percent of responses
30
20
10
0
Q8c
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable

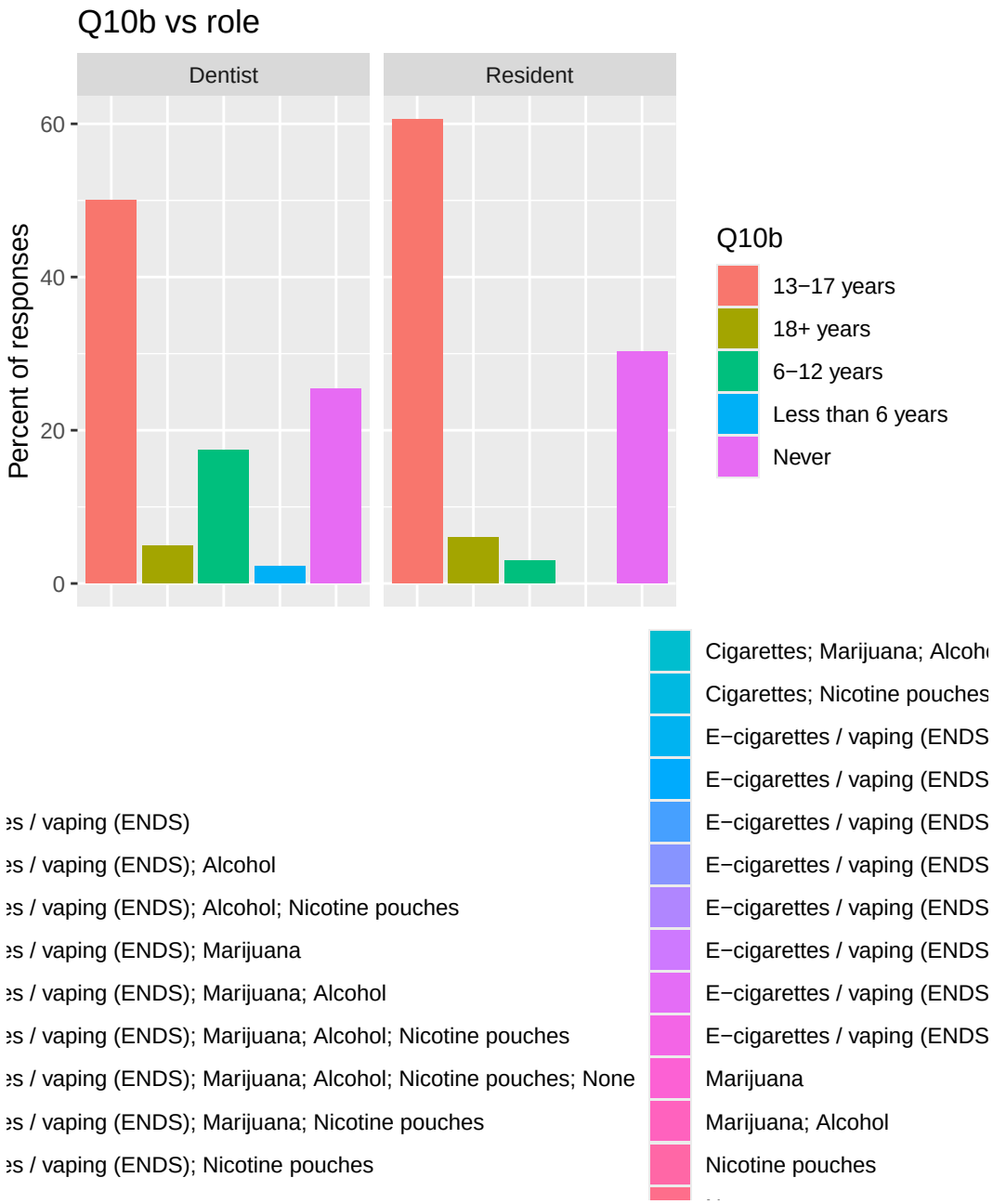

Q8d vs role
Dentist
Resident
Percent of responses
0
10
20
30
40
Q8d
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8e vs role
Dentist
Resident
Percent of responses
0
10
20
30
Q8e
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable

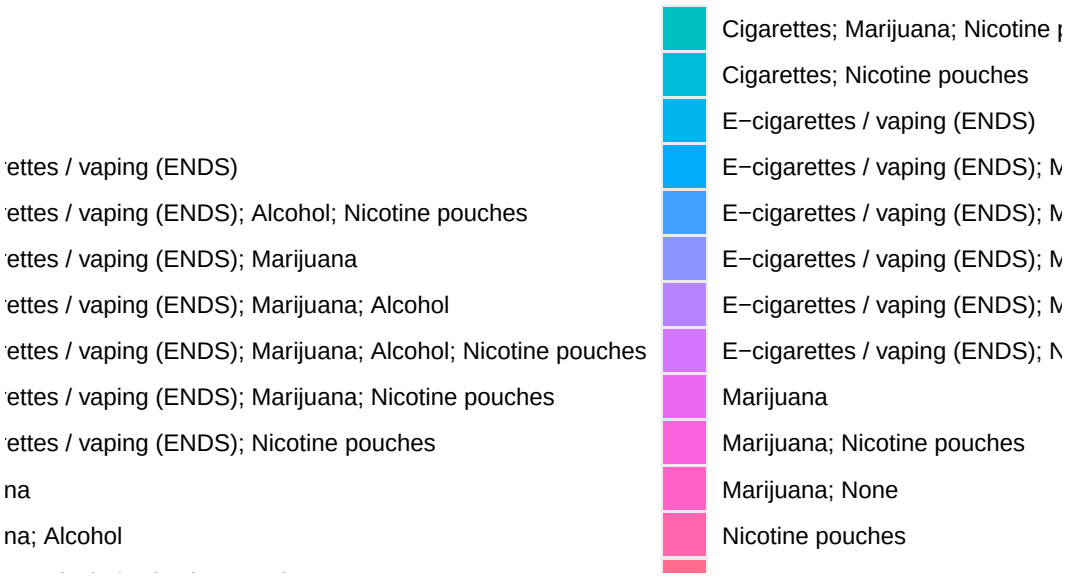

Q8f vs role
Dentist
Resident
Percent of responses
40
30
20
10
0
Q8f
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8g vs role
Dentist
Resident
Percent of responses
40
20
0
Q8g
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable


Q8h vs role
Dentist
Resident
Percent of responses
60
40
20
0
Q8h
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8i vs role
Dentist
Resident
Percent of responses
30
20
10
0
Q8i
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable


Q8j vs role
Dentist
Resident
Percent of responses
30
20
10
0
Q8j
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8k vs role
Dentist
Resident
Percent of responses
60
40
20
0
Q8k
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable

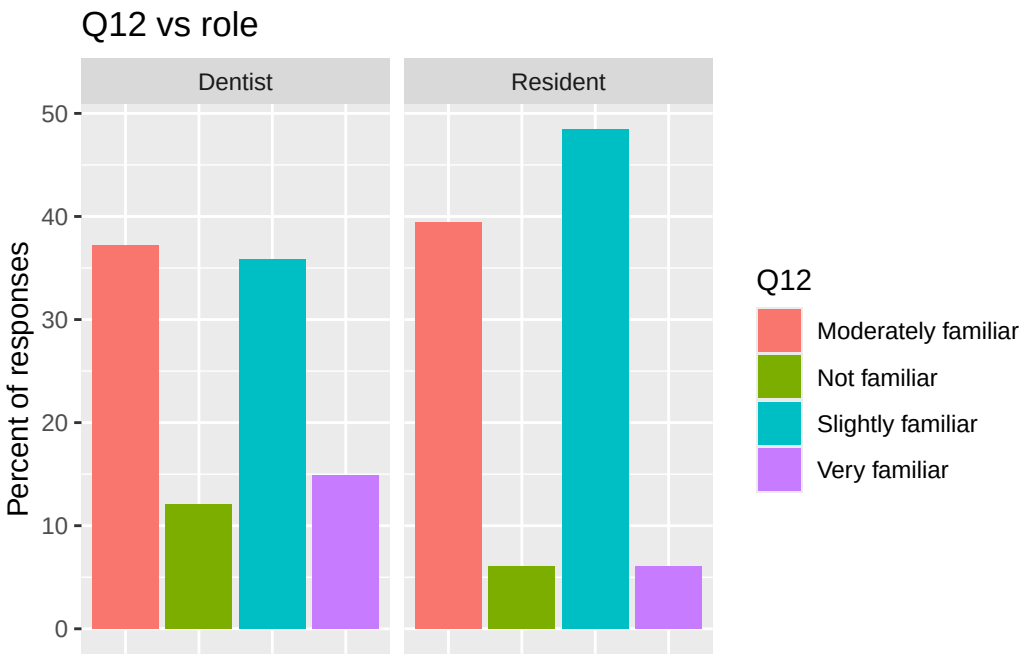

Q9 vs role
Dentist
Resident
Percent of responses
50
40
30
20
10
0
Q9
Always
Never
Often
Rarely
Sometimes
Q10a vs role
Dentist
Resident
Percent of responses
60
40
20
0
Q10a
13−17 years
18+ years
6−12 years
Less than 6 years
Never

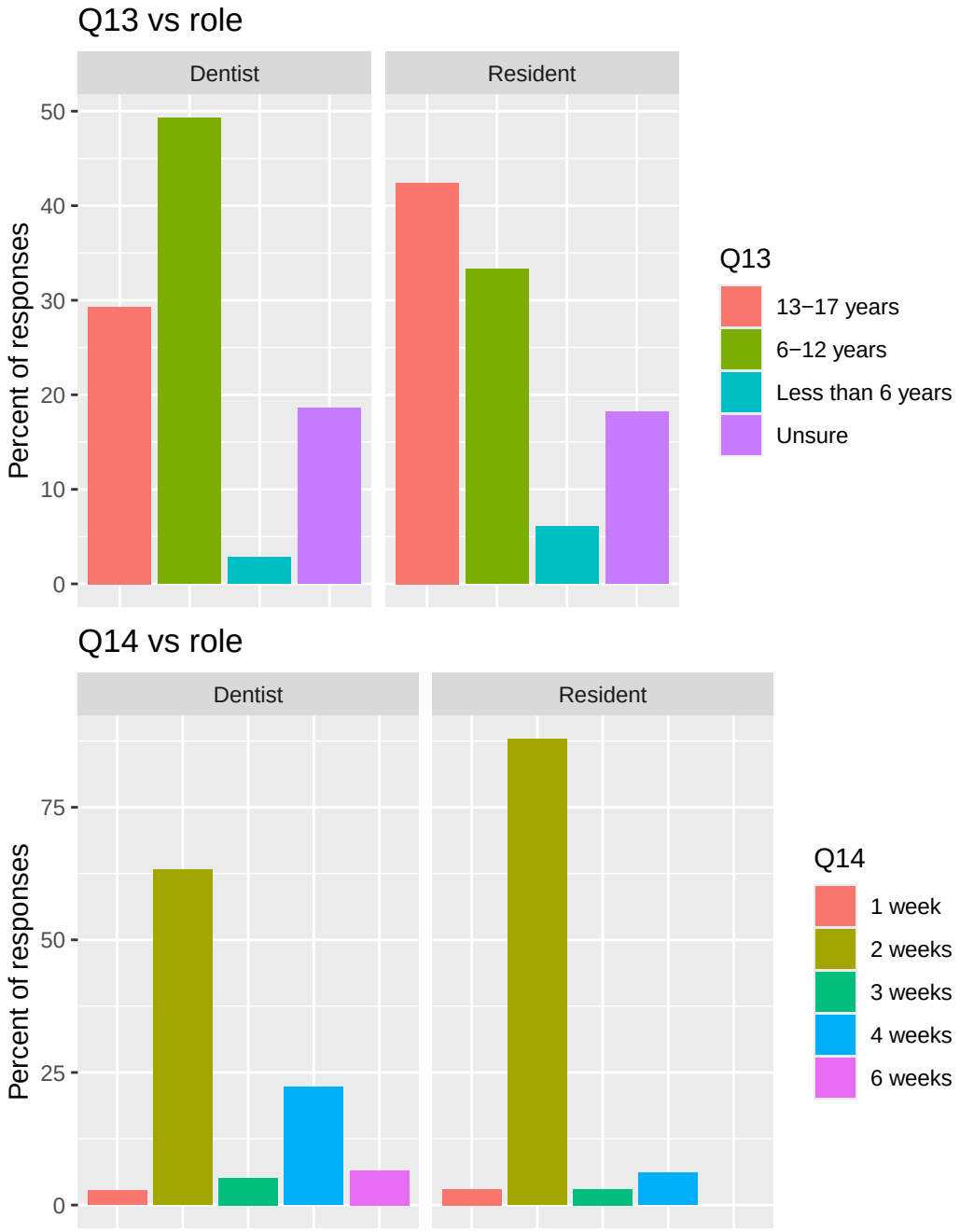

Q10b vs role
Dentist
Resident
Percent of responses
60
40
20
0
Q10b
13−17 years
18+ years
6−12 years
Less than 6 years
Never
es / vaping (ENDS)
es / vaping (ENDS); Alcohol
es / vaping (ENDS); Alcohol; Nicotine pouches
es / vaping (ENDS); Marijuana
es / vaping (ENDS); Marijuana; Alcohol
es / vaping (ENDS); Marijuana; Alcohol; Nicotine pouches
es / vaping (ENDS); Marijuana; Alcohol; Nicotine pouches; None
es / vaping (ENDS); Marijuana; Nicotine pouches
es / vaping (ENDS); Nicotine pouches
Cigarettes; Marijuana; Alcoh
Cigarettes; Nicotine pouches
E−cigarettes / vaping (ENDS
E−cigarettes / vaping (ENDS
E−cigarettes / vaping (ENDS
E−cigarettes / vaping (ENDS
E−cigarettes / vaping (ENDS
E−cigarettes / vaping (ENDS
E−cigarettes / vaping (ENDS
Marijuana
Marijuana; Alcohol
Nicotine pouches



Knowledge Plots

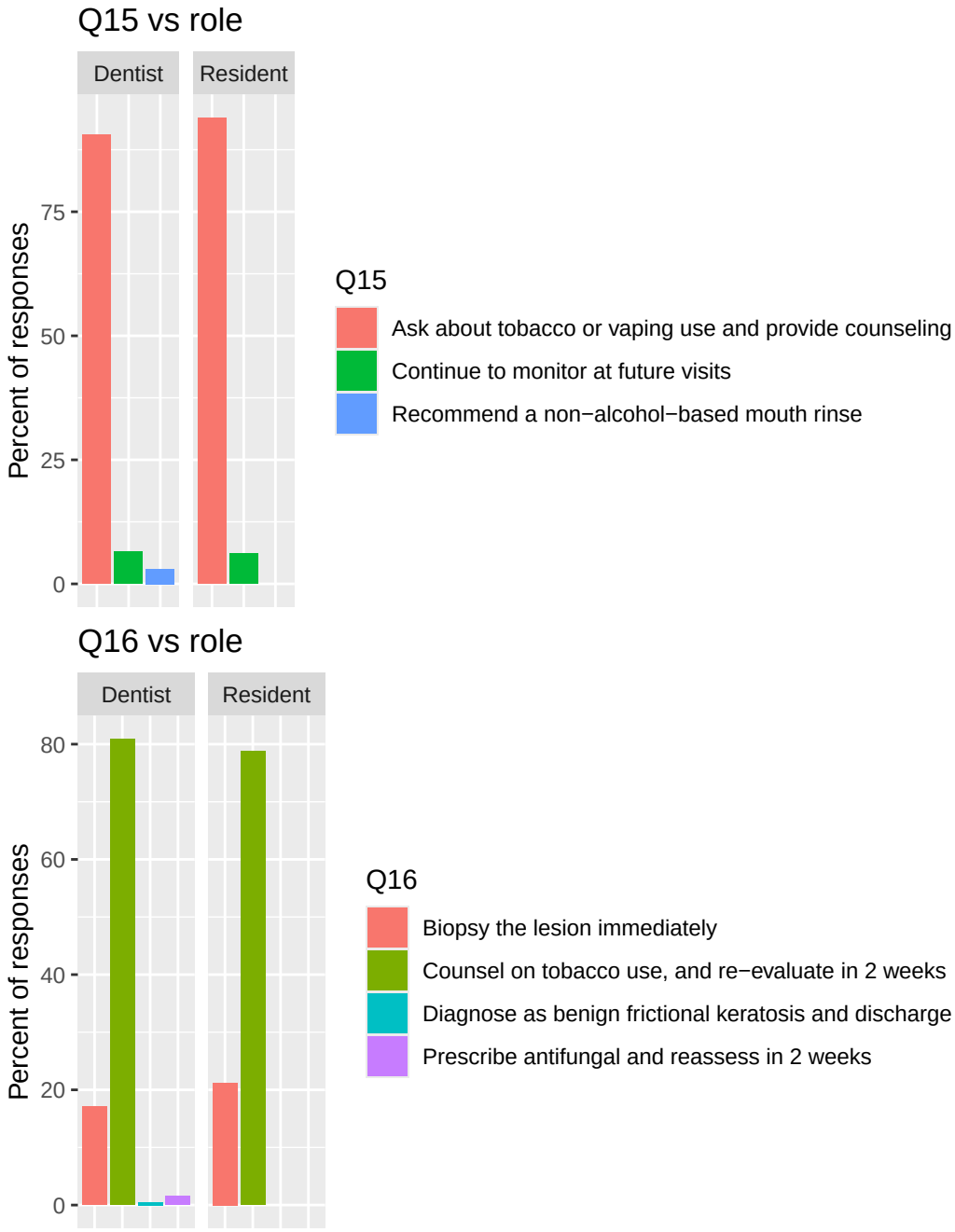
Below are plots comparing the knowledge of residents versus dentists.

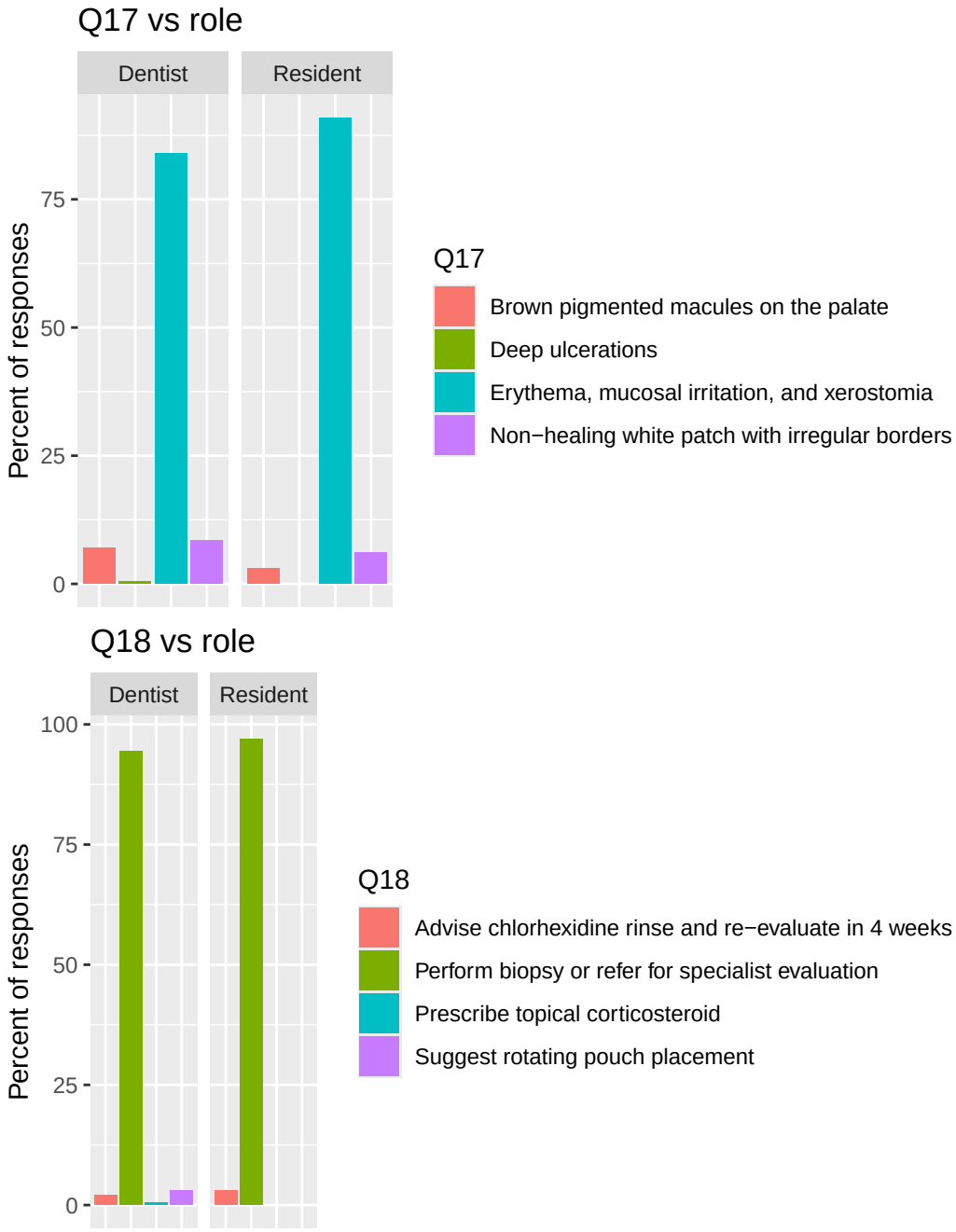


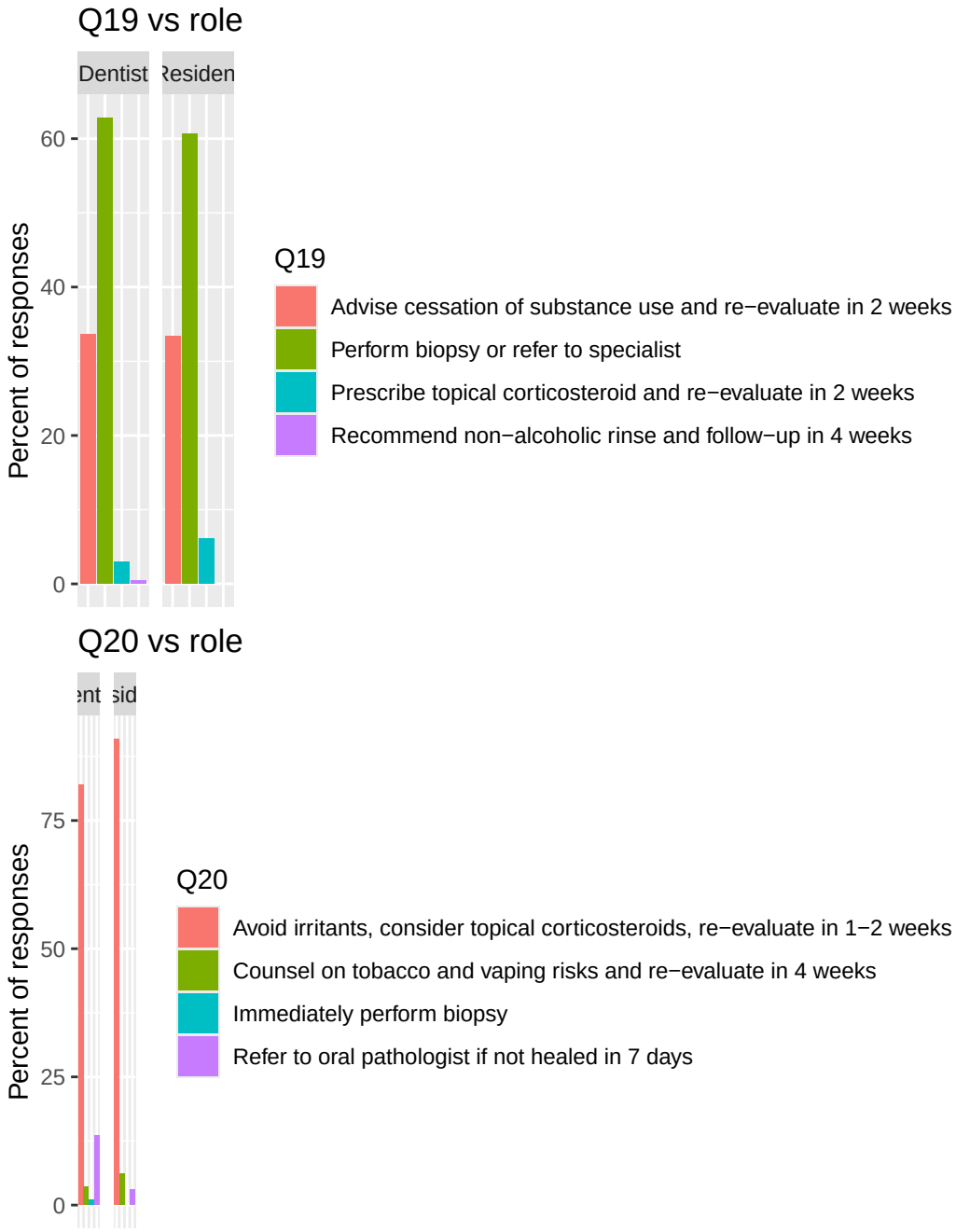


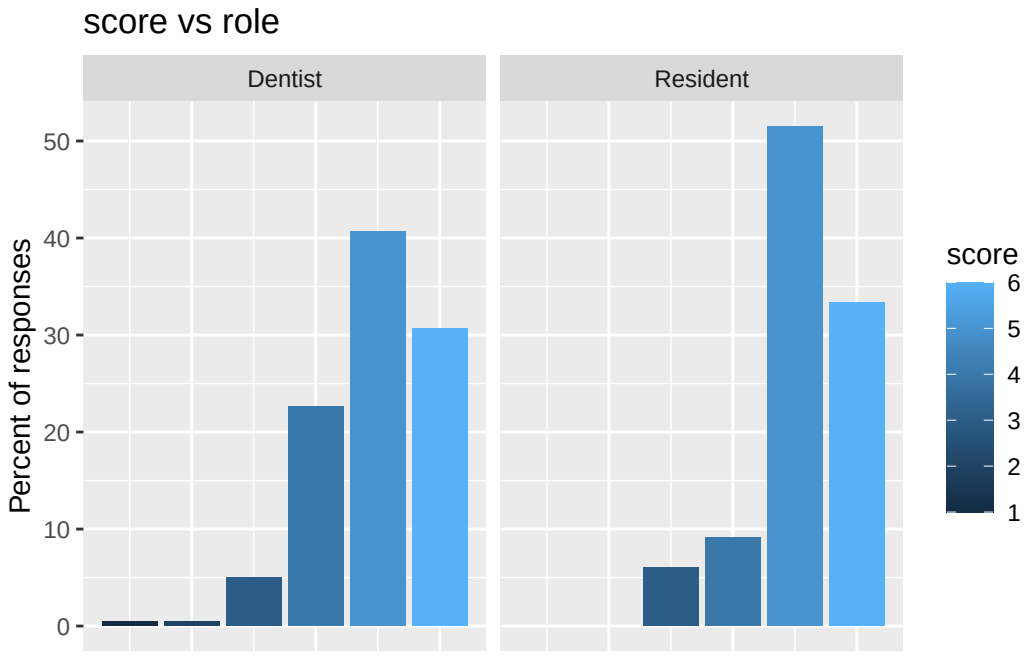
Quiz Plots

Below are plots comparing the quiz results between residents and dentists.


Q15 vs role
Dentist
Resident
Percent of responses
75
50
25
0
Q15
Ask about tobacco or vaping use and provide counseling
Continue to monitor at future visits
Recommend a non−alcohol−based mouth rinse
Q16 vs role
Dentist
Resident
Percent of responses
80
60
40
20
0
Q16
Biopsy the lesion immediately
Counsel on tobacco use, and re−evaluate in 2 weeks
Diagnose as benign frictional keratosis and discharge
Prescribe antifungal and reassess in 2 weeks

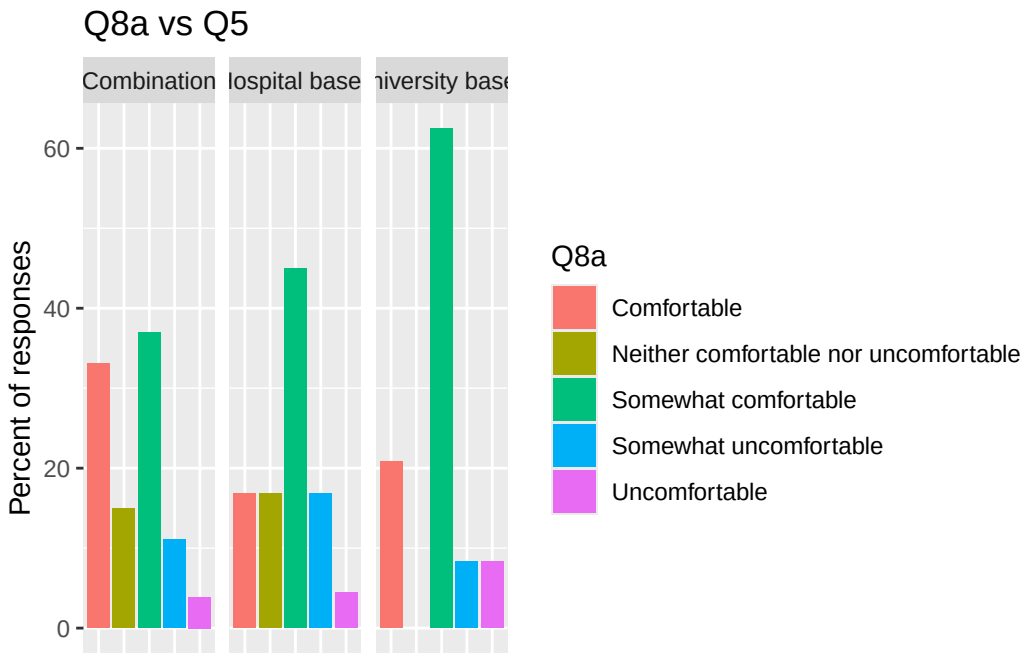

Q17 vs role
Dentist
Resident
Percent of responses
75
50
25
0
Q17
Brown pigmented macules on the palate
Deep ulcerations
Erythema, mucosal irritation, and xerostomia
Non−healing white patch with irregular borders
Q18 vs role
Dentist
Resident
100
75
50
25
0
Percent of responses
Q18
Advise chlorhexidine rinse and re−evaluate in 4 weeks
Perform biopsy or refer for specialist evaluation
Prescribe topical corticosteroid
Suggest rotating pouch placement

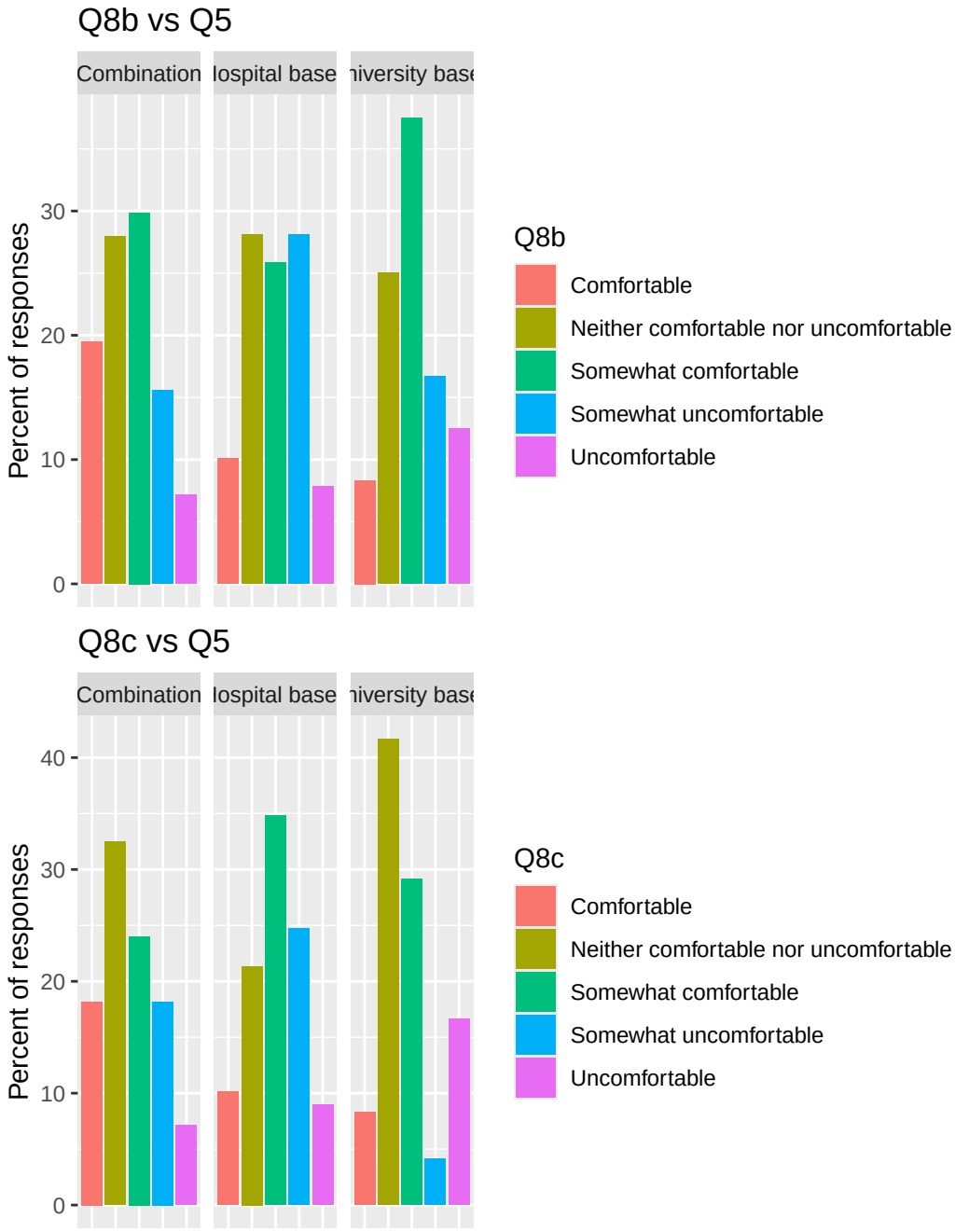

Q19 vs role
Dentist
Residen
Percent of responses
60
40
20
0
Q19
Advise cessation of substance use and re−evaluate in 2 weeks
Perform biopsy or refer to specialist
Prescribe topical corticosteroid and re−evaluate in 2 weeks
Recommend non−alcoholic rinse and follow−up in 4 weeks
Q20 vs role
Percent of responses
75
50
25
0
Q20
Avoid irritants, consider topical corticosteroids, re−evaluate in 1−2 weeks
Counsel on tobacco and vaping risks and re−evaluate in 4 weeks
Immediately perform biopsy
Refer to oral pathologist if not healed in 7 days

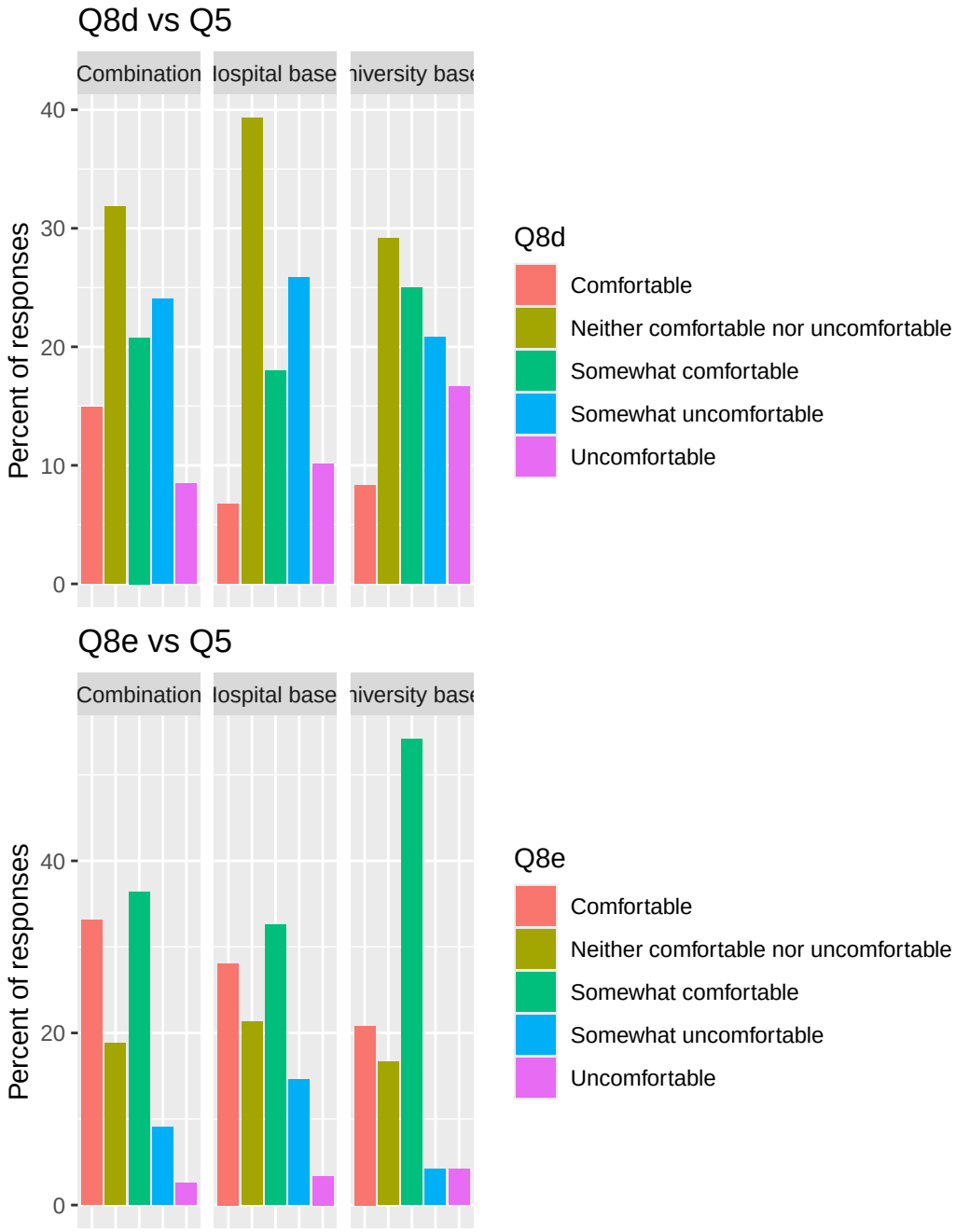


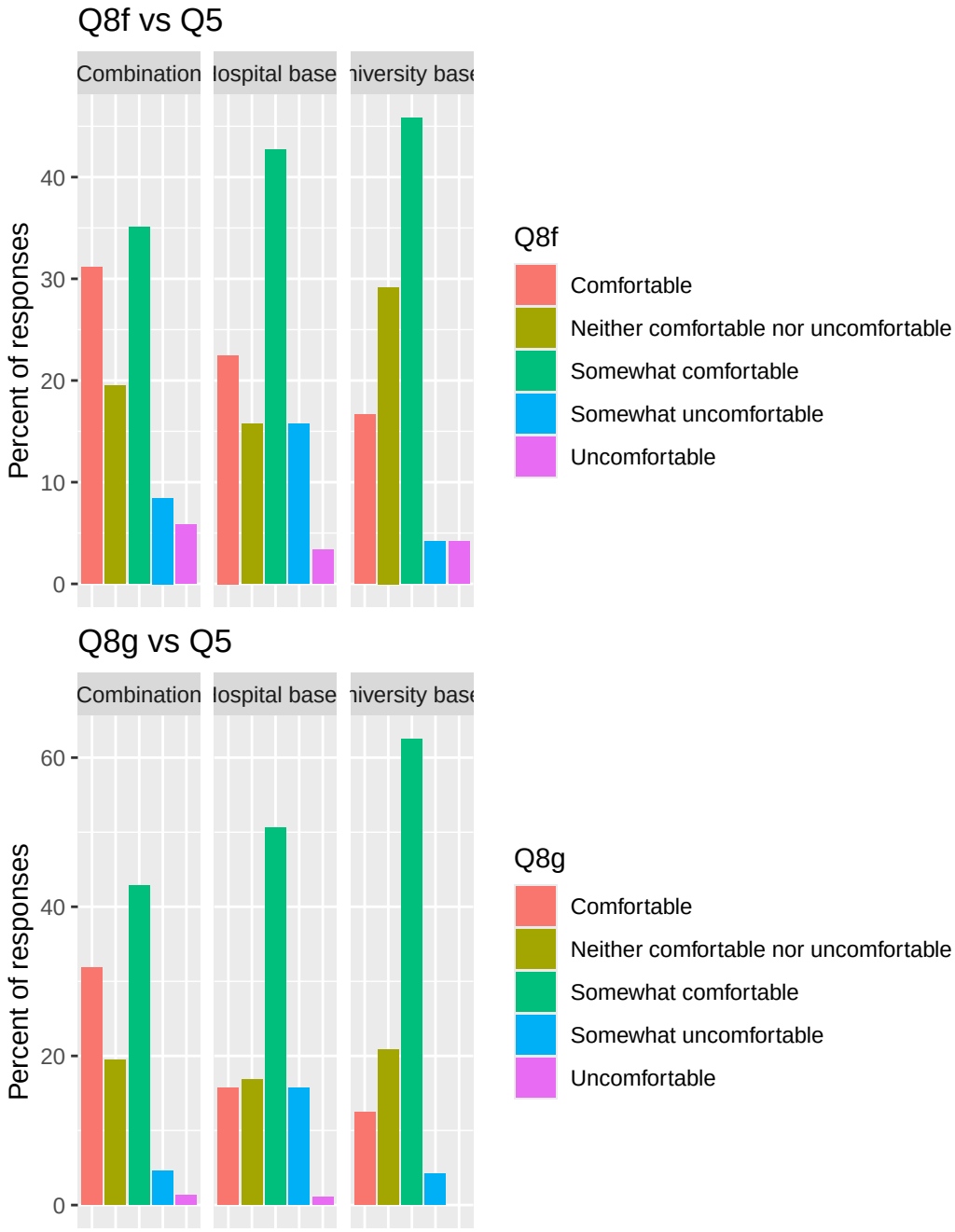
Environmental Analysis Plots

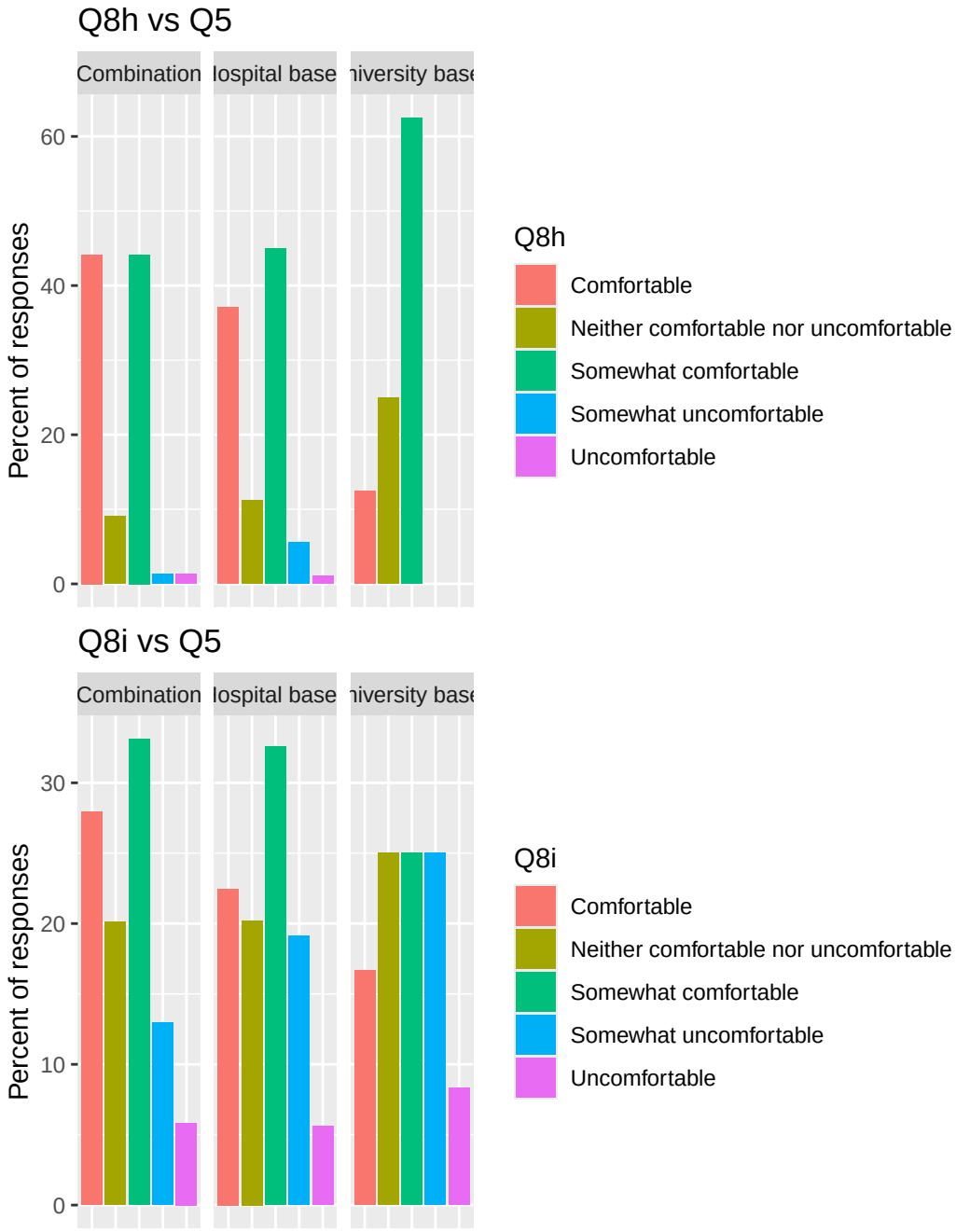
Below are plots for the environmental analysis.

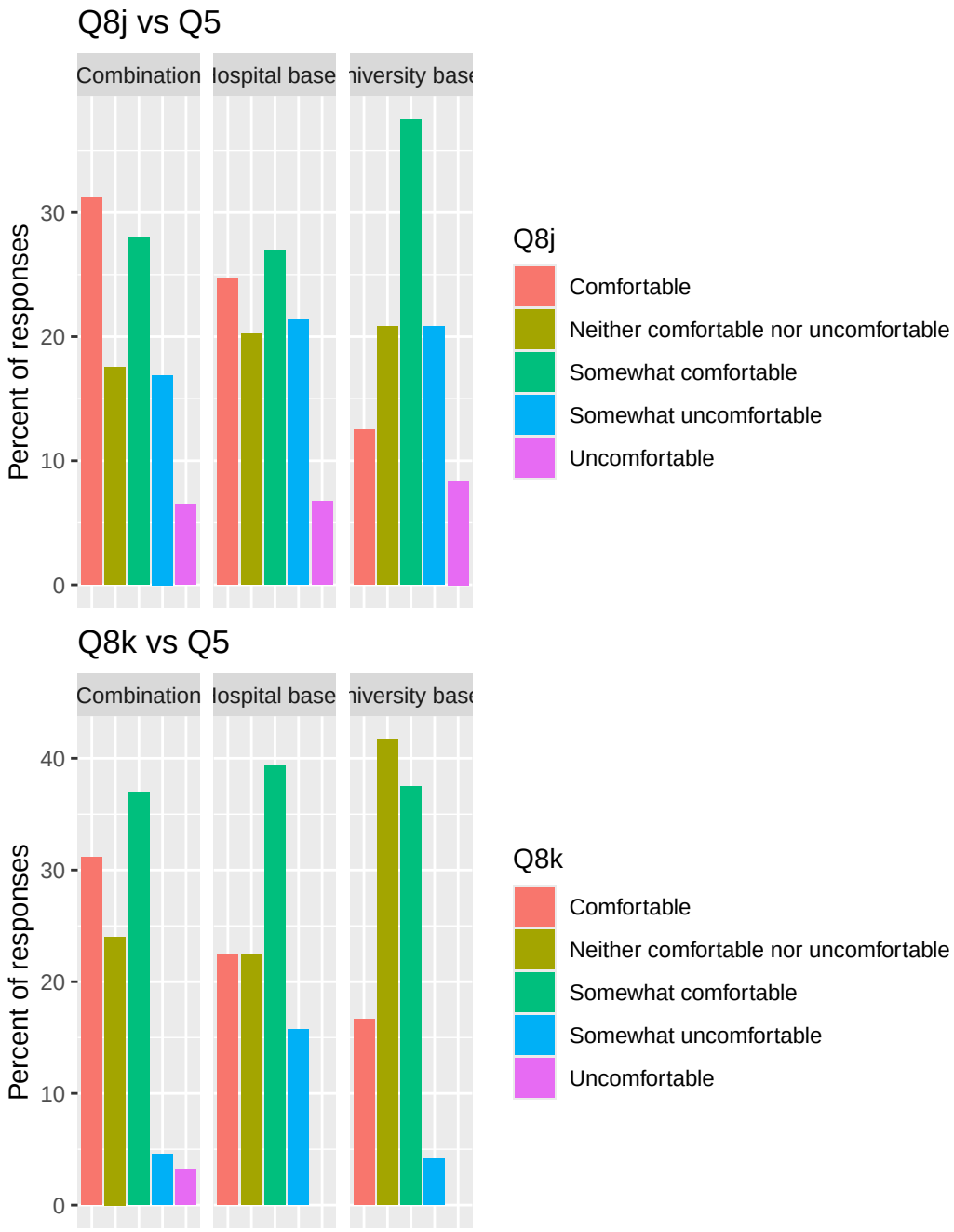


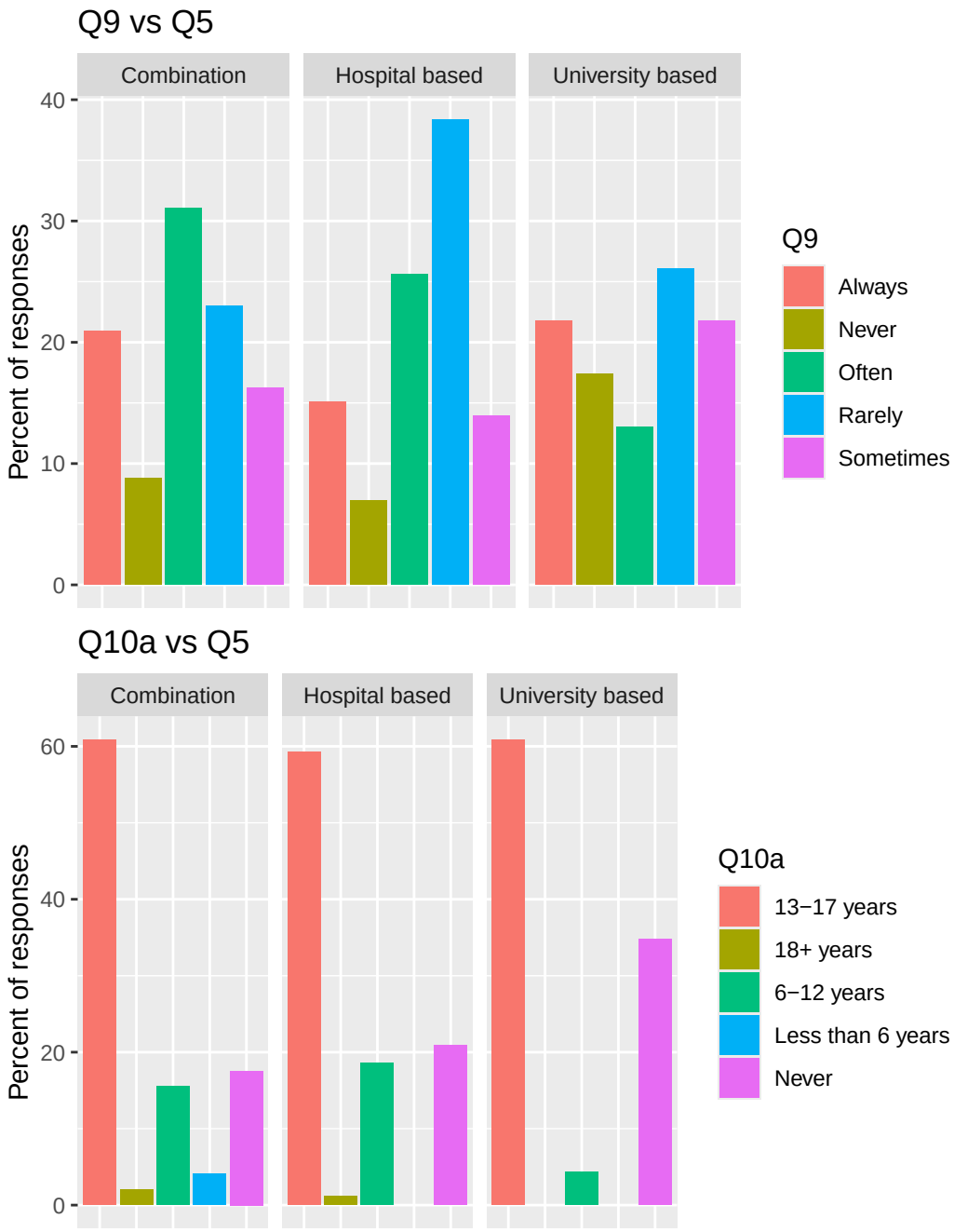

Q8b vs Q5
Combination
lospital base
niversity base
Percent of responses
0
10
20
30
Q8b
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8c vs Q5
Combination
lospital base
niversity base
Percent of responses
0
10
20
30
40
Q8c
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable

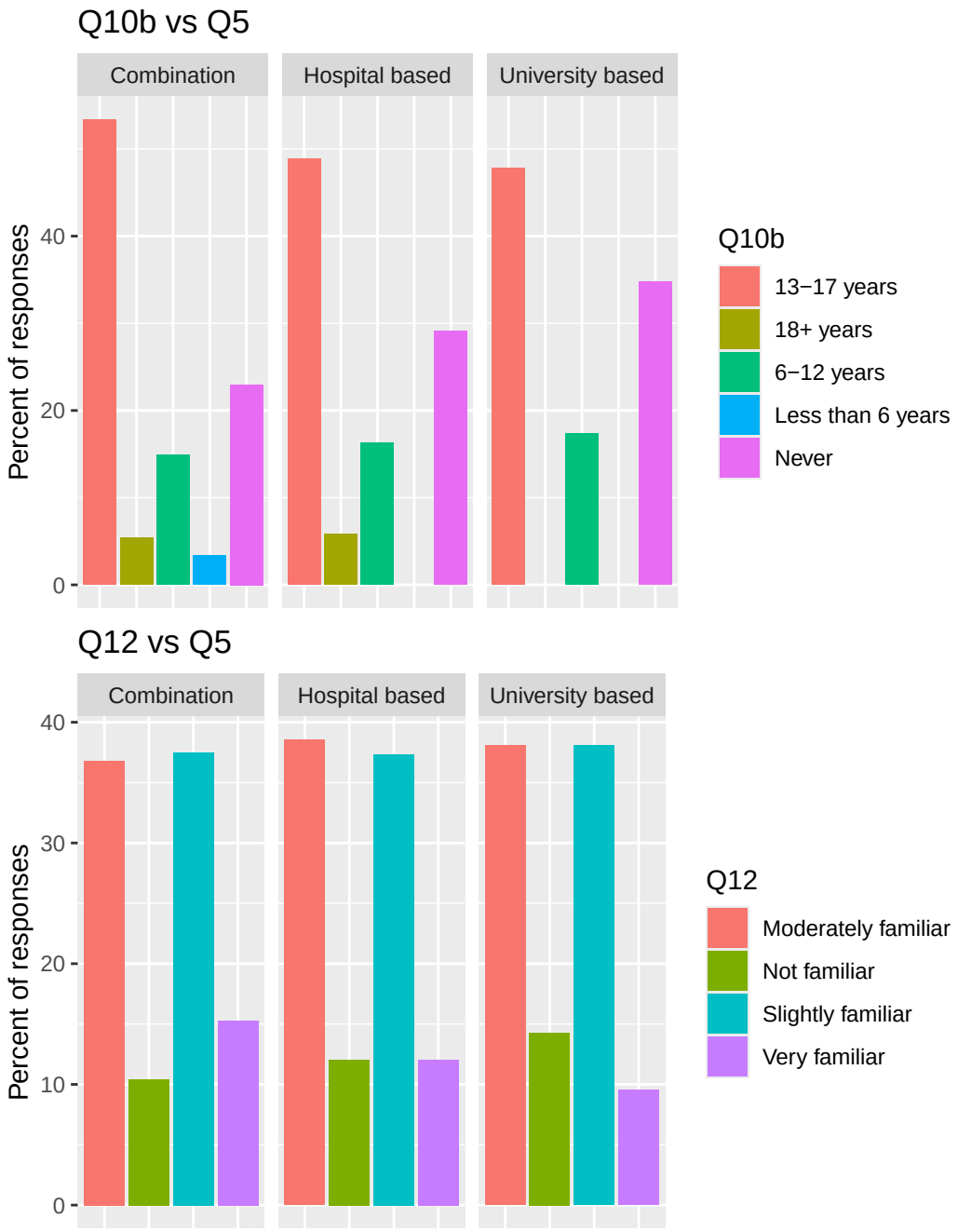

Q8d vs Q5
Combination
lospital base
niversity base
Percent of responses
40
30
20
10
0
Q8d
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8e vs Q5
Combination
lospital base
niversity base
Percent of responses
40
20
0
Q8e
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable

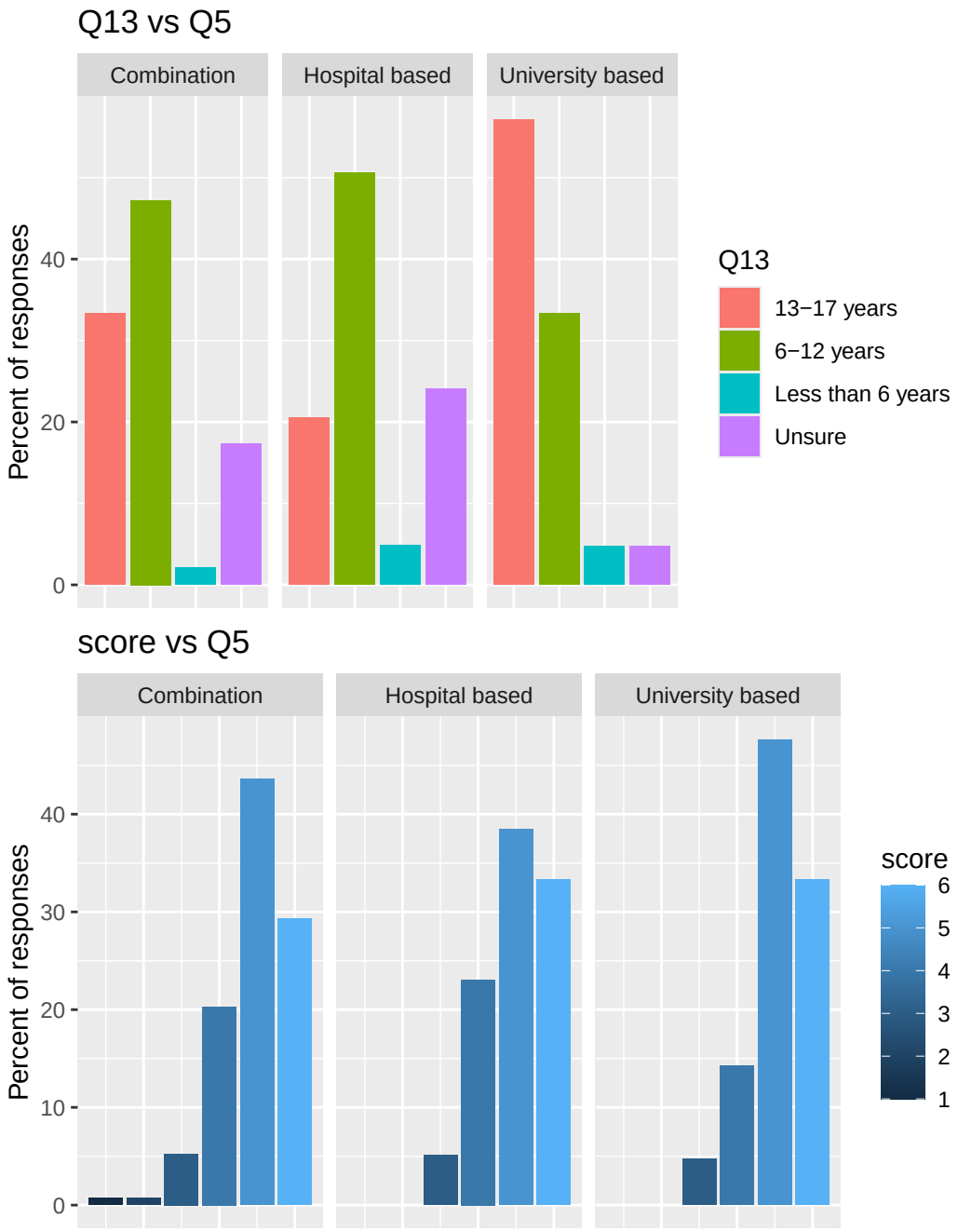

Q8f vs Q5
Combination
lospital base
niversity base
Percent of responses
0
10
20
30
40
Q8f
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8g vs Q5
Combination
lospital base
niversity base
Percent of responses
0
20
40
60
Q8g
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable


Q8h vs Q5
Combination
lospital base
niversity base
Percent of responses
60
40
20
0
Q8h
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8i vs Q5
Combination
lospital base
niversity base
Percent of responses
30
20
10
0
Q8i
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable


Q8j vs Q5
Combination
lospital base
niversity base
Percent of responses
0
10
20
30
Q8j
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8k vs Q5
Combination
lospital base
niversity base
Percent of responses
0
10
20
30
40
Q8k
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable

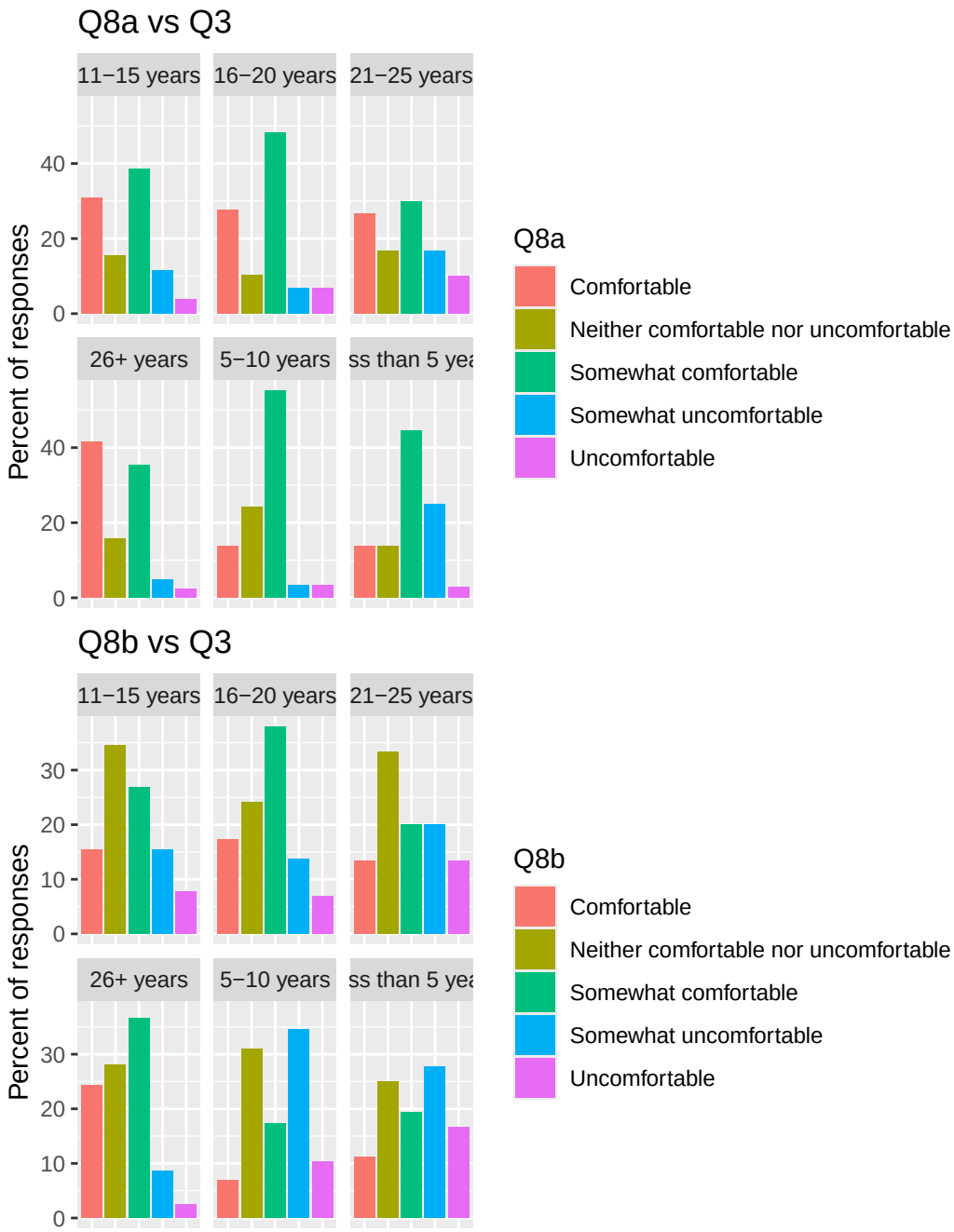

Q9 vs Q5
Combination
Hospital based
University based
Percent of responses
0
10
20
30
40
Q9
Always
Never
Often
Rarely
Sometimes
Q10a vs Q5
Combination
Hospital based
University based
Percent of responses
0
20
40
60
Q10a
13−17 years
18+ years
6−12 years
Less than 6 years
Never

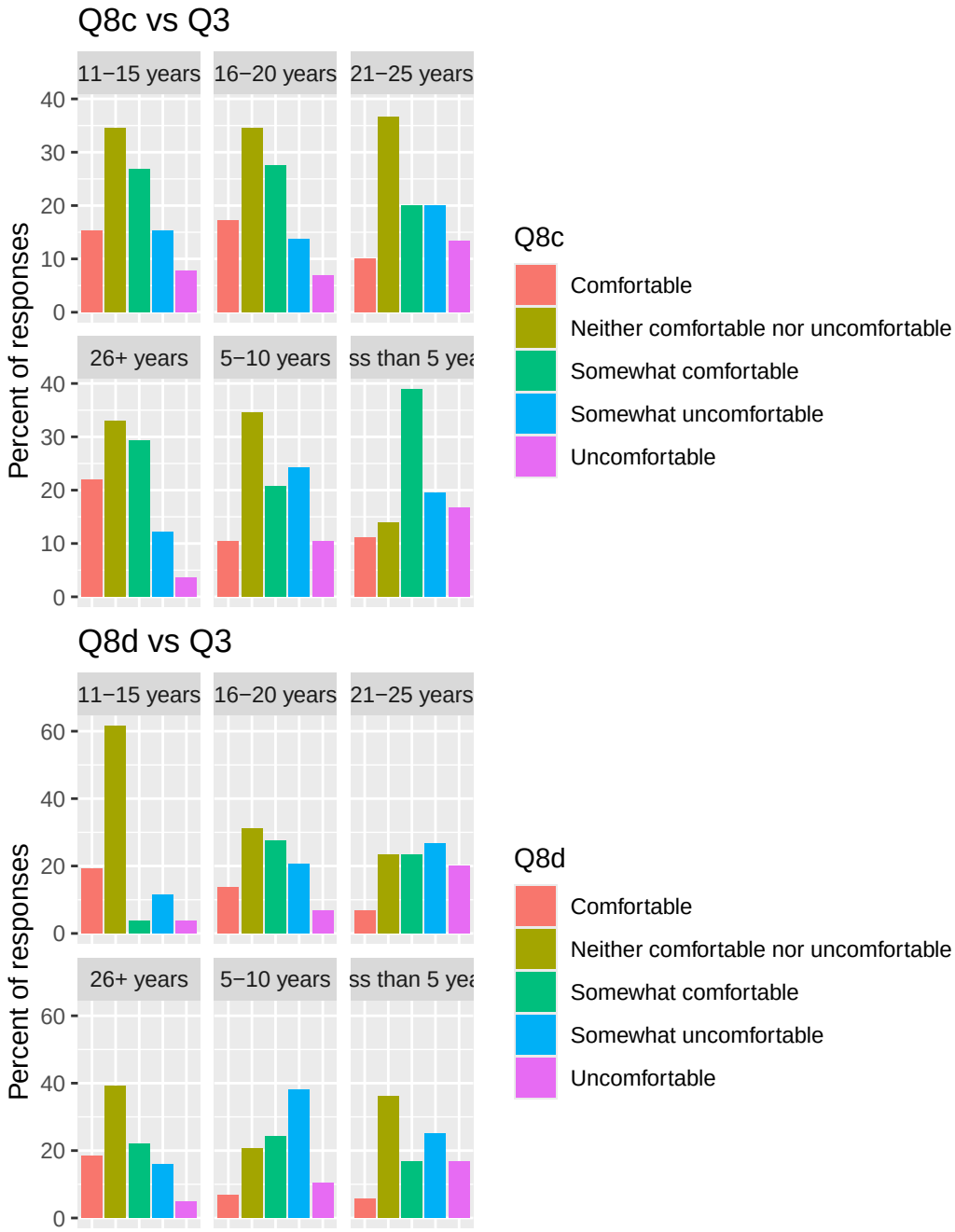

Q10b vs Q5
Combination
Hospital based
University based
Percent of responses
40
20
0
Q10b
13−17 years
18+ years
6−12 years
Less than 6 years
Never
Q12 vs Q5
Combination
Hospital based
University based
Percent of responses
40
30
20
10
0
Q12
Moderately familiar
Not familiar
Slightly familiar
Very familiar

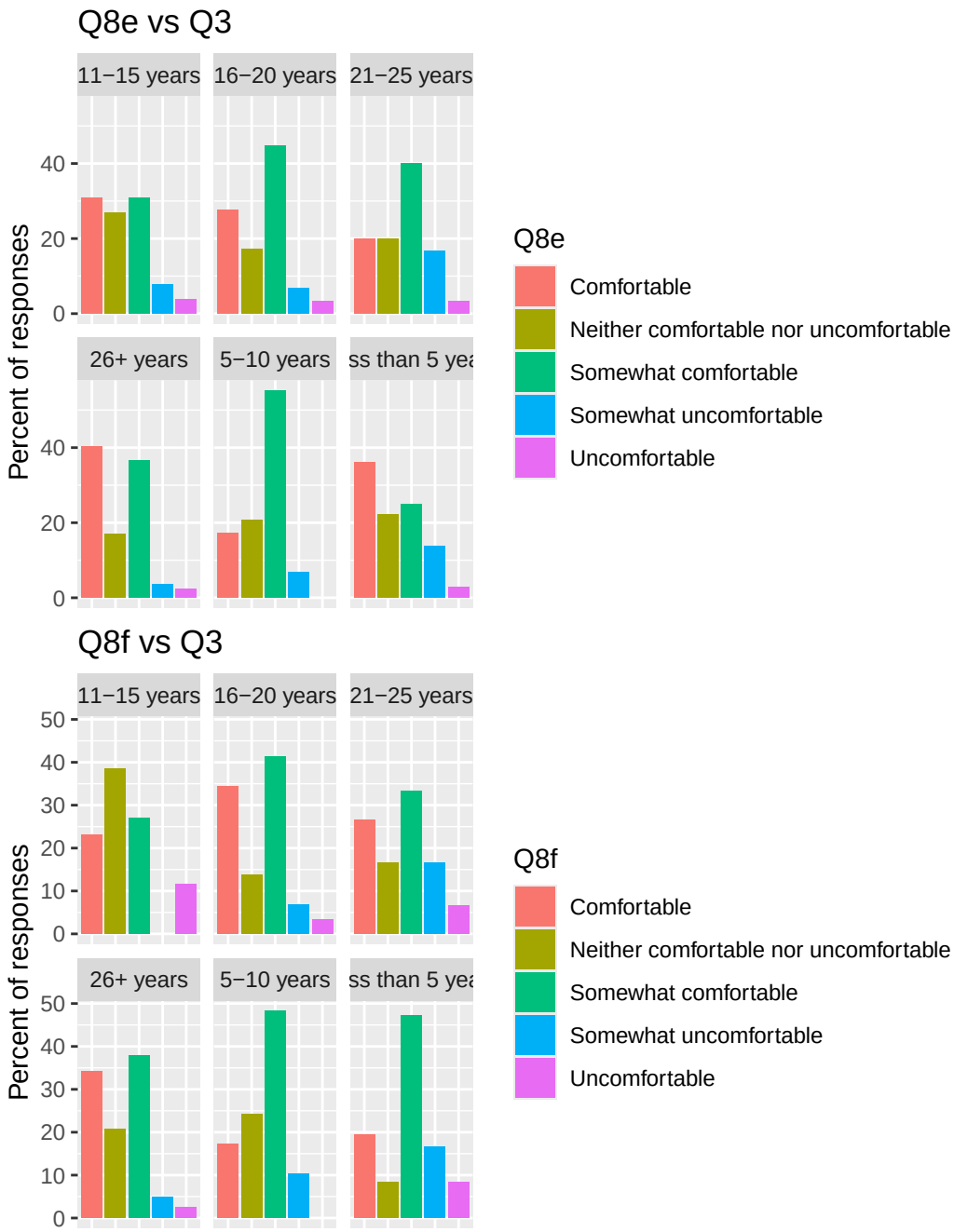


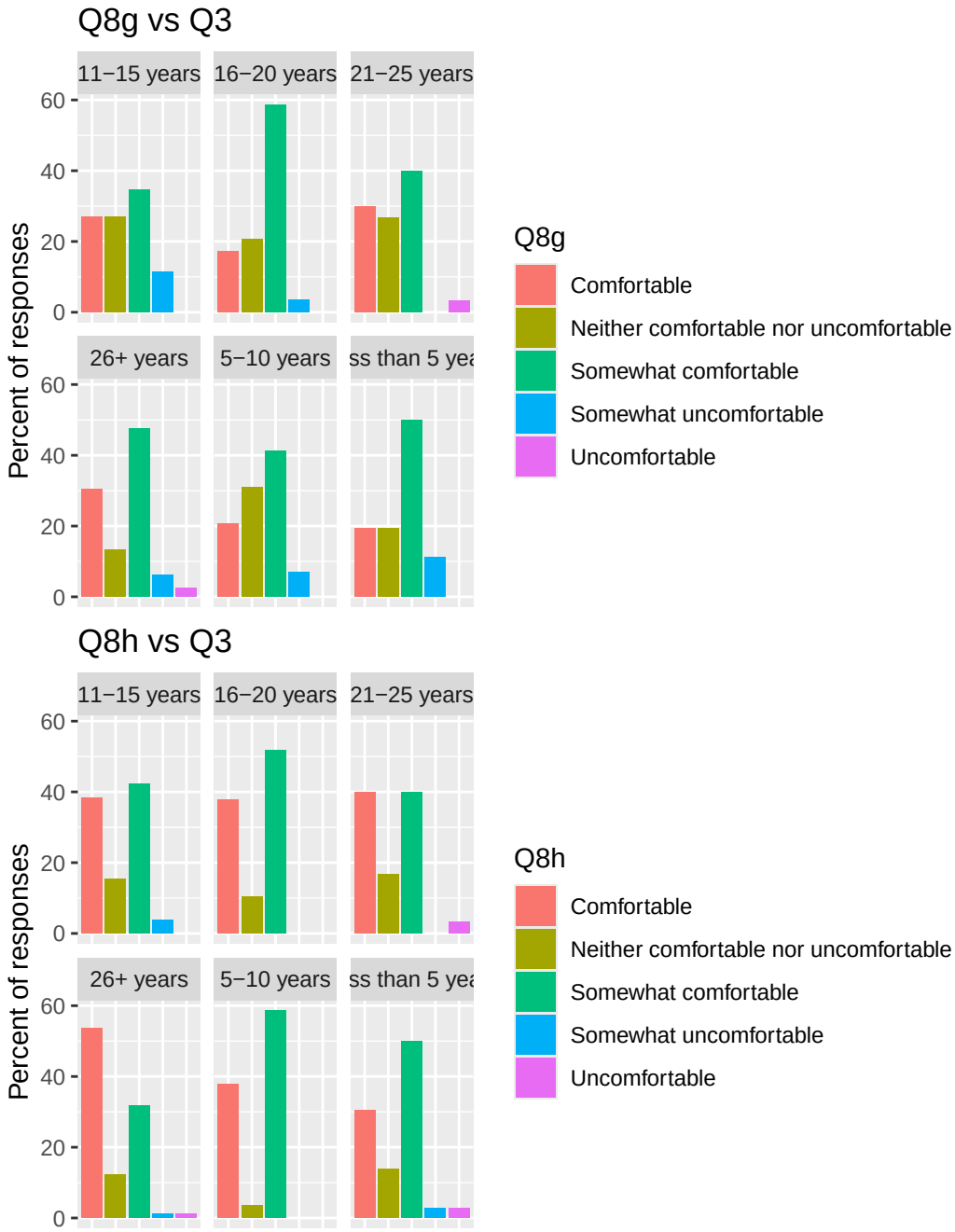
Dentist Analysis Plots

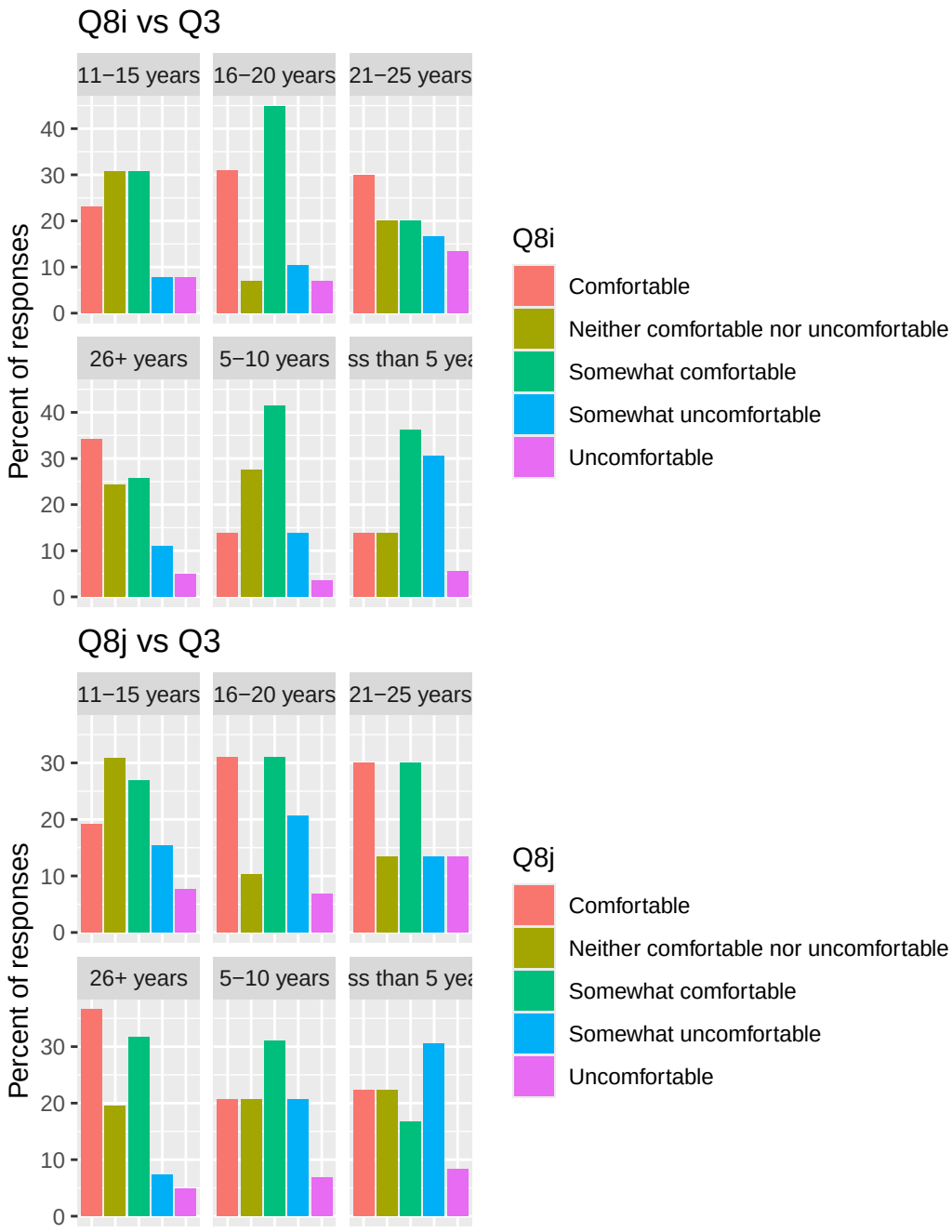
Below are plots for the dentist analysis.

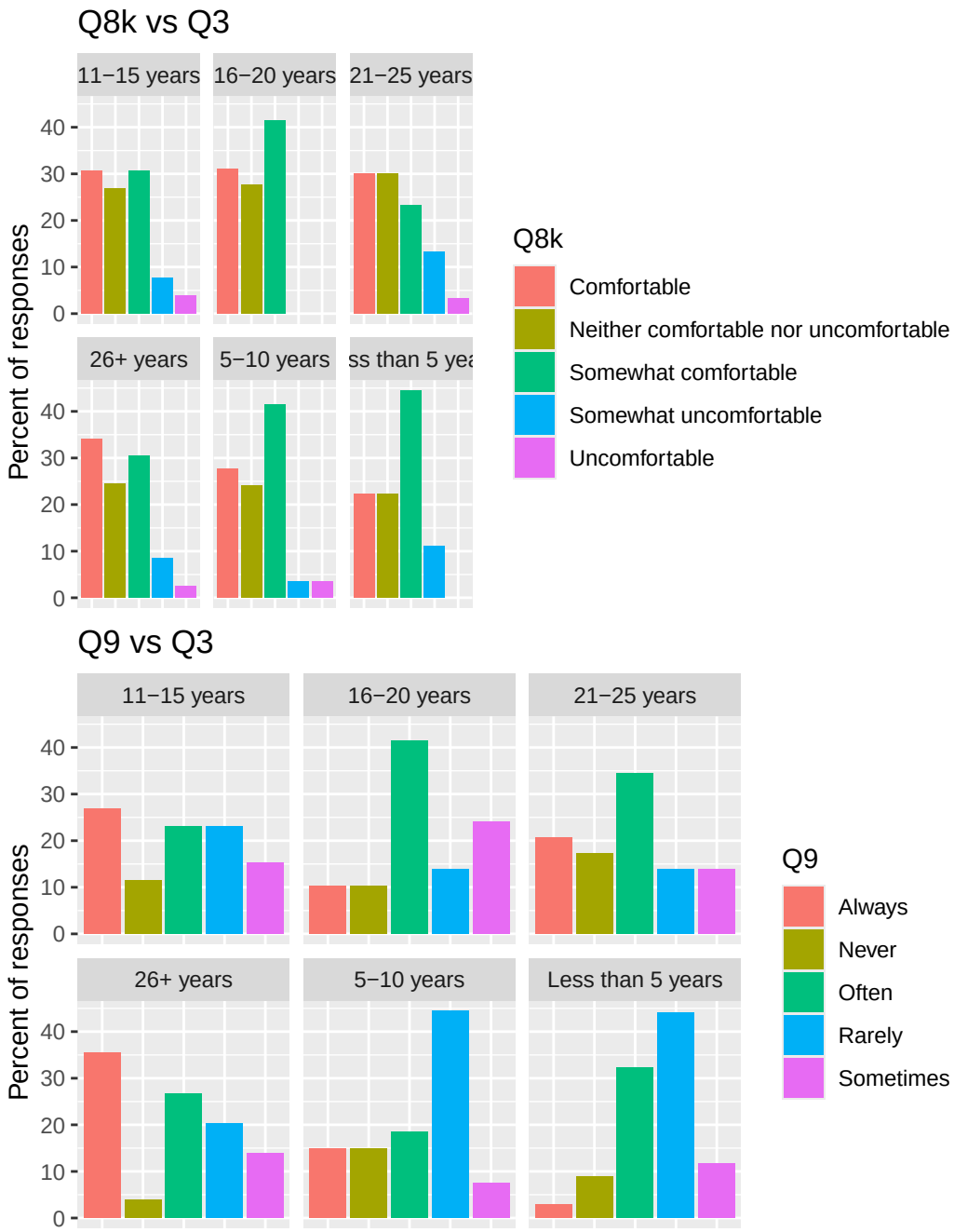

Q8a vs Q3
11−15 years
16−20 years
21−25 years
26+ years
5−10 years
ss than 5 yea
Percent of responses
0
20
40
Q8a
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8b vs Q3
11−15 years
16−20 years
21−25 years
26+ years
5−10 years
ss than 5 yea
Percent of responses
0
10
20
30
Q8b
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable

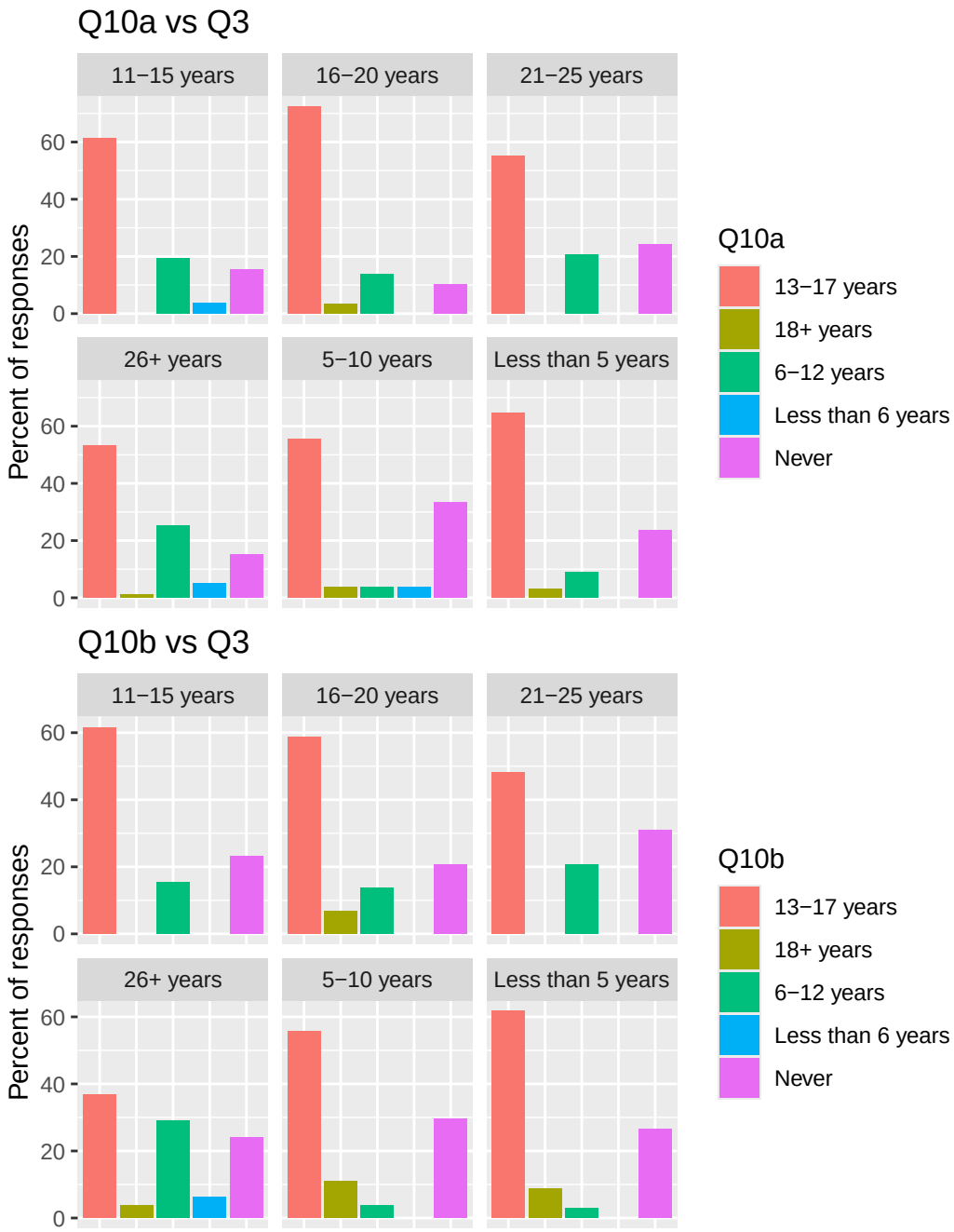

Q8c vs Q3
11−15 years
16−20 years
21−25 years
26+ years
5−10 years
ss than 5 yea
Percent of responses
0
10
20
30
40
Q8c
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8d vs Q3
11−15 years
16−20 years
21−25 years
26+ years
5−10 years
ss than 5 yea
Percent of responses
0
20
40
60
Q8d
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable


Q8e vs Q3
11−15 years
16−20 years
21−25 years
26+ years
5−10 years
ss than 5 yea
Percent of responses
0
20
40
Q8e
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8f vs Q3
11−15 years
16−20 years
21−25 years
26+ years
5−10 years
ss than 5 yea
Percent of responses
0
10
20
30
40
50
Q8f
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable

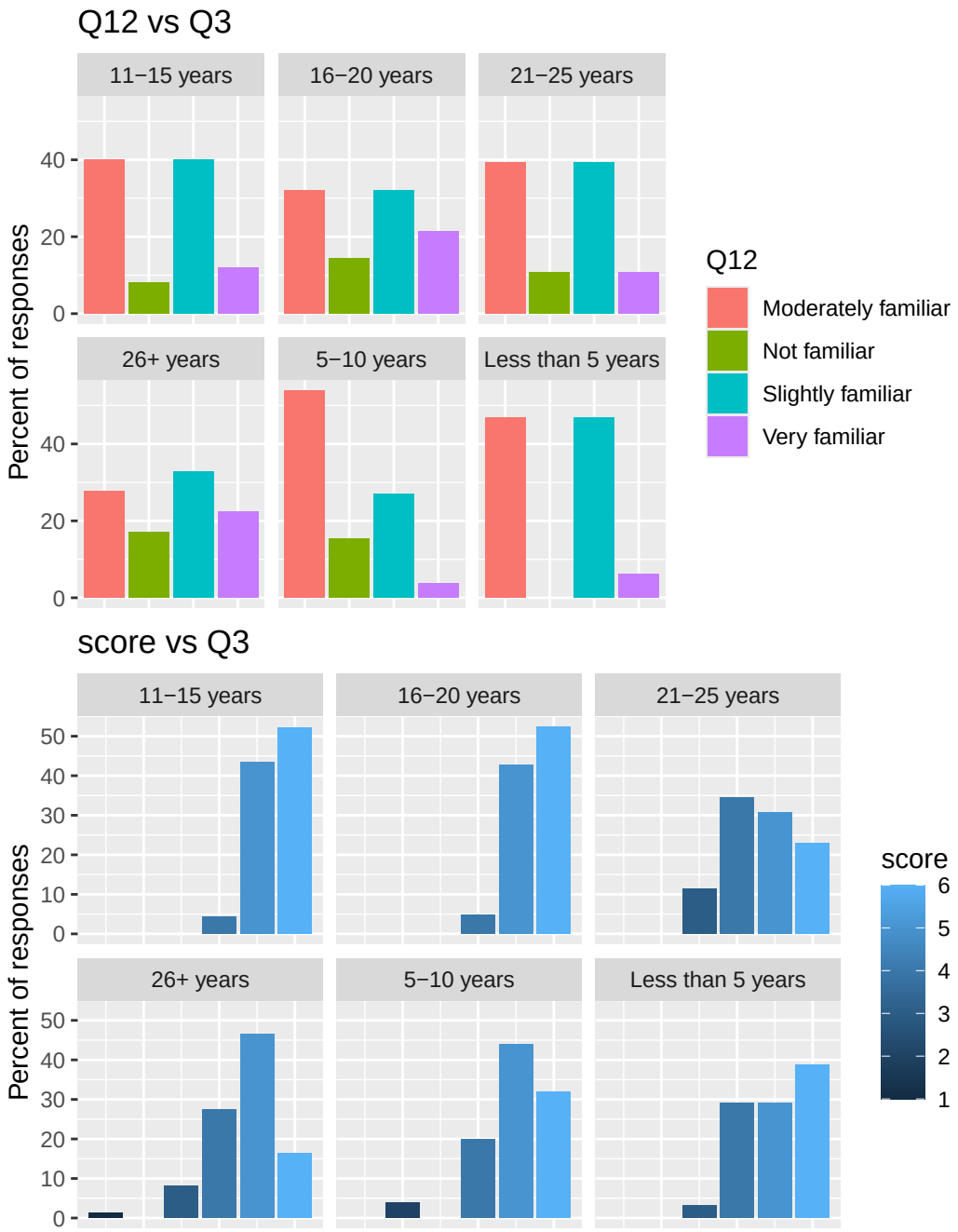

Q8g vs Q3
11−15 years
16−20 years
21−25 years
26+ years
5−10 years
ss than 5 yea
Percent of responses
0
20
40
60
Q8g
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8h vs Q3
11−15 years
16−20 years
21−25 years
26+ years
5−10 years
ss than 5 yea
Percent of responses
0
20
40
60
Q8h
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable


Q8i vs Q3
11−15 years
16−20 years
21−25 years
26+ years
5−10 years
ss than 5 yea
Percent of responses
0
10
20
30
40
Q8i
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8j vs Q3
11−15 years
16−20 years
21−25 years
26+ years
5−10 years
ss than 5 yea
Percent of responses
0
10
20
30
Q8j
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable


Q8k vs Q3
11−15 years
16−20 years
21−25 years
26+ years
5−10 years
ss than 5 yea
Percent of responses
0
10
20
30
40
Q8k
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q9 vs Q3
11−15 years
16−20 years
21−25 years
26+ years
5−10 years
Less than 5 years
Percent of responses
0
10
20
30
40
Q9
Always
Never
Often
Rarely
Sometimes

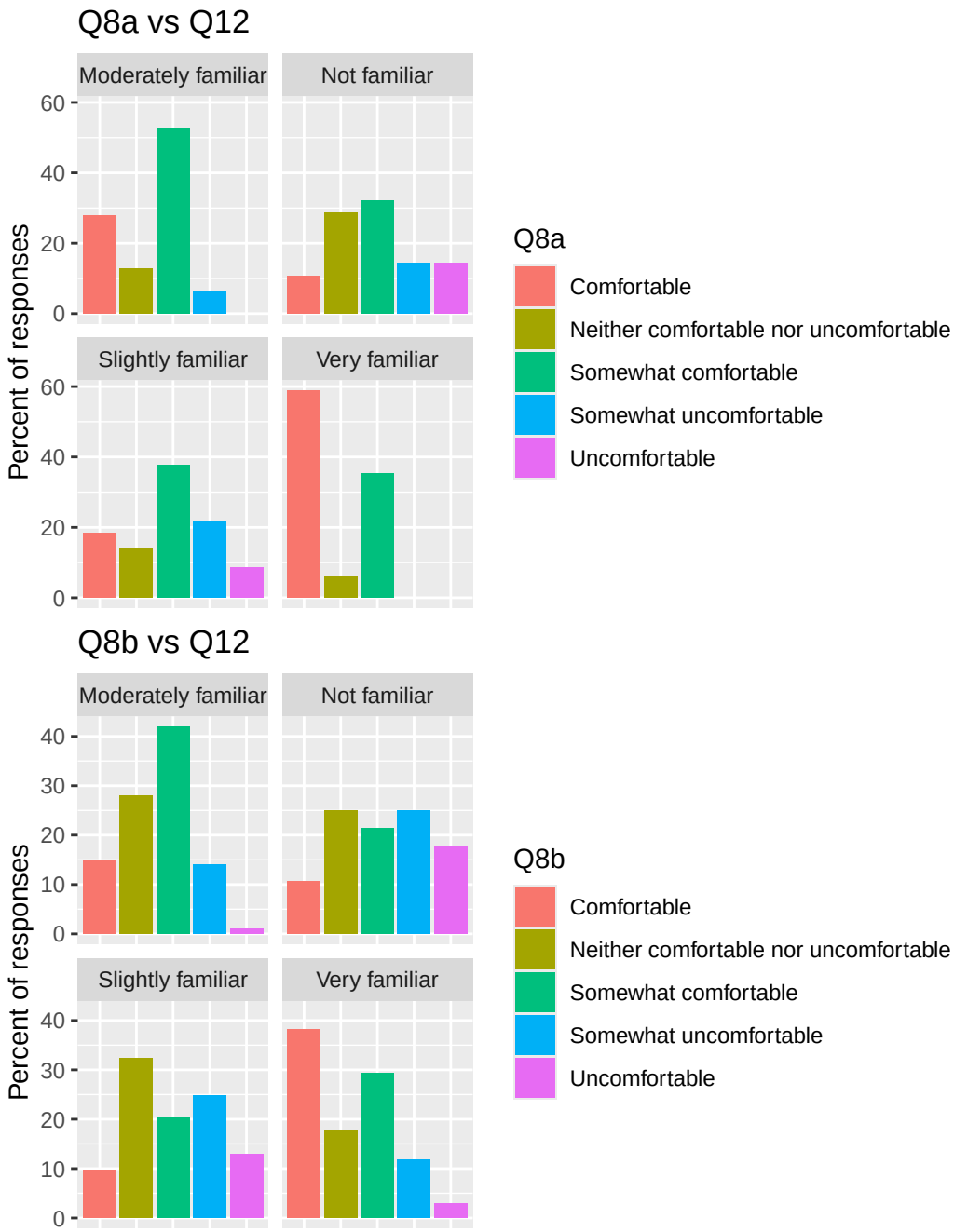

Q10a vs Q3
11−15 years
16−20 years
21−25 years
26+ years
5−10 years
Less than 5 years
Percent of responses
0
20
40
60
Q10a
13−17 years
18+ years
6−12 years
Less than 6 years
Never
Q10b vs Q3
11−15 years
16−20 years
21−25 years
26+ years
5−10 years
Less than 5 years
Percent of responses
0
20
40
60
Q10b
13−17 years
18+ years
6−12 years
Less than 6 years
Never

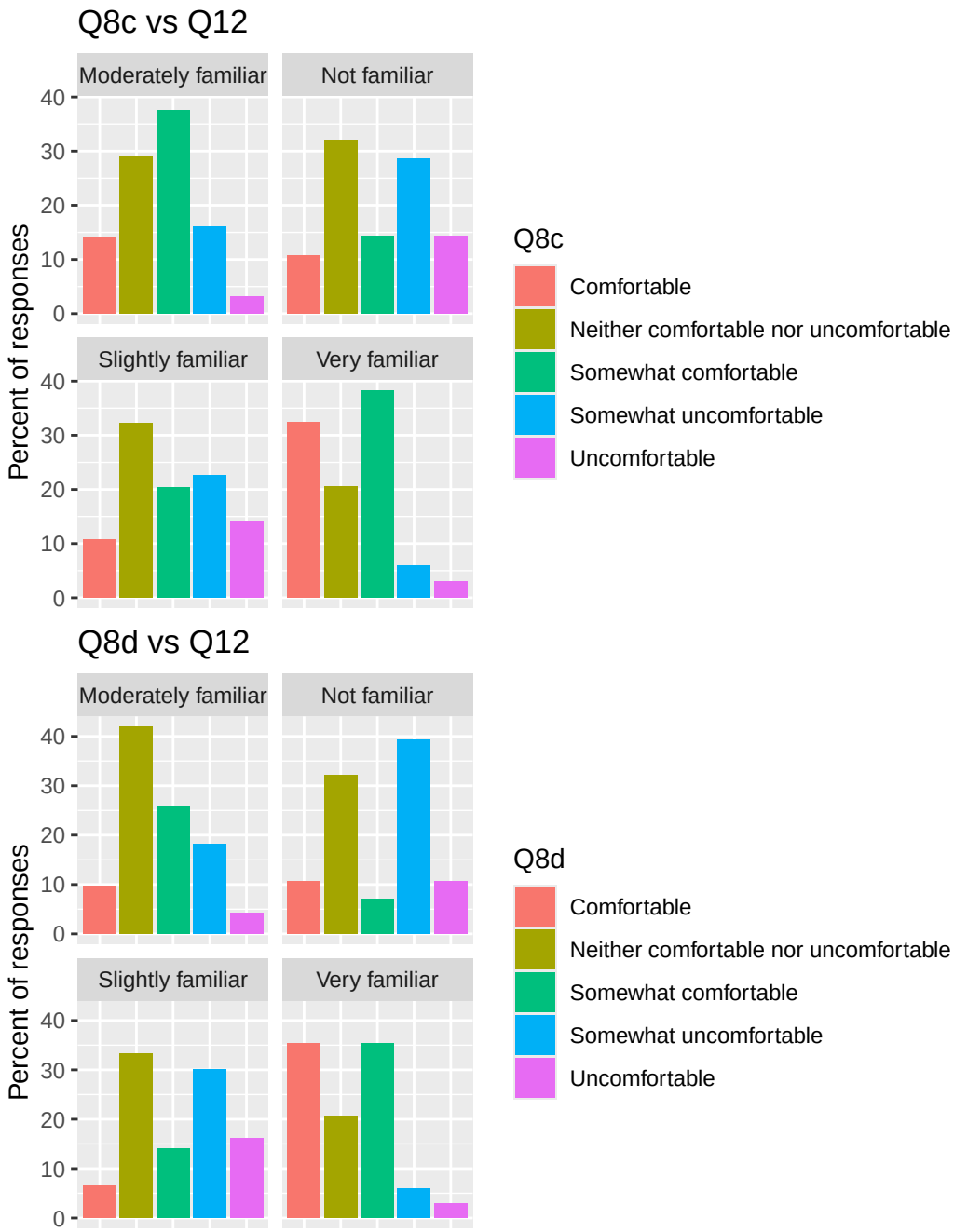

es / vaping (ENDS)
es / vaping (ENDS); Alcohol
es / vaping (ENDS); Alcohol; Nicotine pouches
es / vaping (ENDS); Marijuana
es / vaping (ENDS); Marijuana; Alcohol
es / vaping (ENDS); Marijuana; Alcohol; Nicotine pouches
es / vaping (ENDS); Marijuana; Alcohol; Nicotine pouches; None
es / vaping (ENDS); Marijuana; Nicotine pouches
es / vaping (ENDS); Nicotine pouches
Cigarettes; Marijuana; Alcoh
Cigarettes; Nicotine pouches
E−cigarettes / vaping (ENDS
E−cigarettes / vaping (ENDS
E−cigarettes / vaping (ENDS
E−cigarettes / vaping (ENDS
E−cigarettes / vaping (ENDS
E−cigarettes / vaping (ENDS
E−cigarettes / vaping (ENDS
Marijuana
Marijuana; Alcohol
Nicotine pouches
:ohol
·cigarettes / vaping (ENDS)
·cigarettes / vaping (ENDS); Alcohol; Nicotine pouches
·cigarettes / vaping (ENDS); Marijuana
·cigarettes / vaping (ENDS); Marijuana; Alcohol
·cigarettes / vaping (ENDS); Marijuana; Alcohol; Nicotine pouches
·cigarettes / vaping (ENDS); Marijuana; Nicotine pouches
·cigarettes / vaping (ENDS); Nicotine pouches
ɪrijuana
ɪrijuana; Alcohol
Cigarettes; Marijuana; Nicc
Cigarettes; Nicotine pouche
E−cigarettes / vaping (END
E−cigarettes / vaping (END
E−cigarettes / vaping (END
E−cigarettes / vaping (END
E−cigarettes / vaping (END
Marijuana
Marijuana; Nicotine pouche
Marijuana; None
Nicotine pouches
None

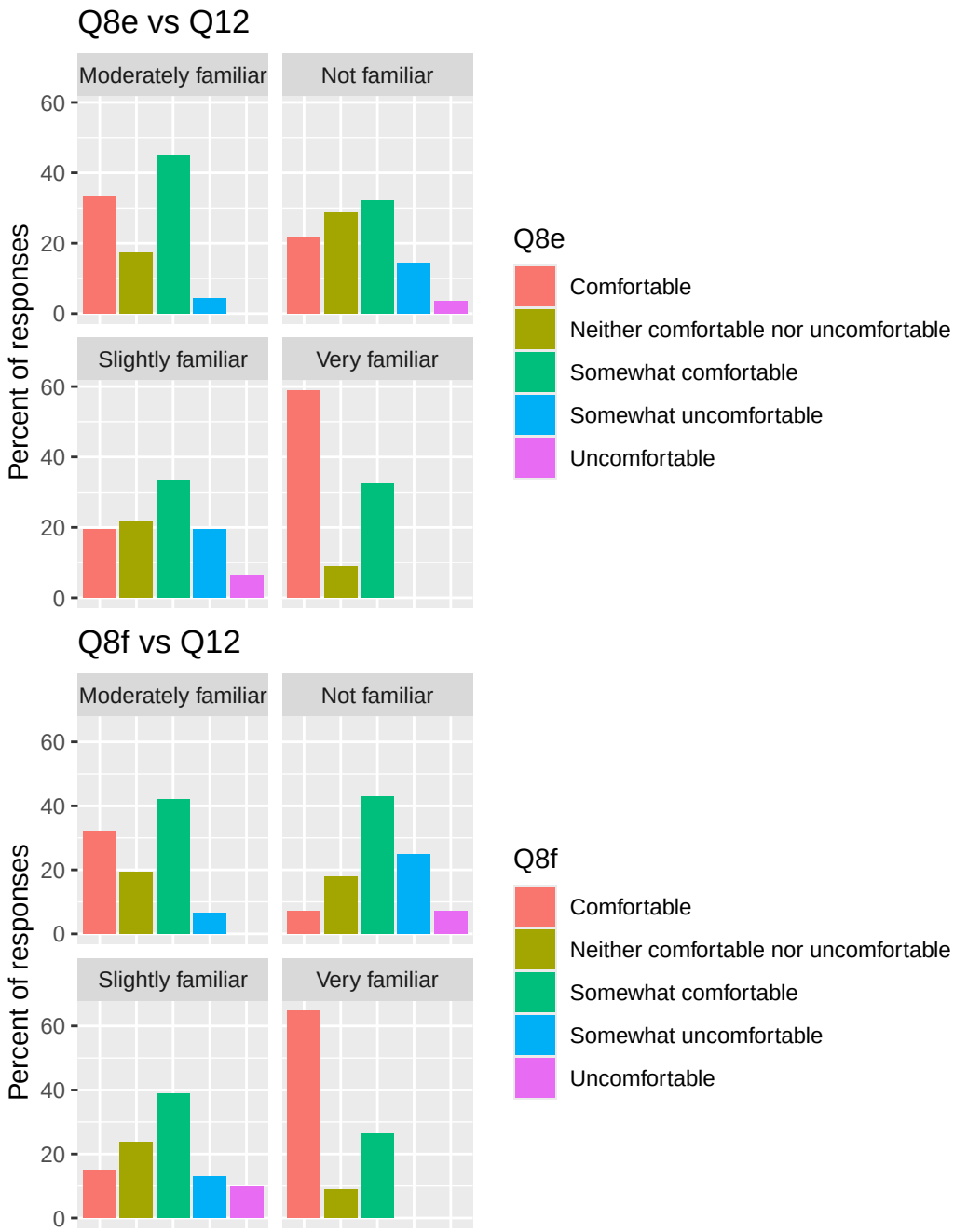


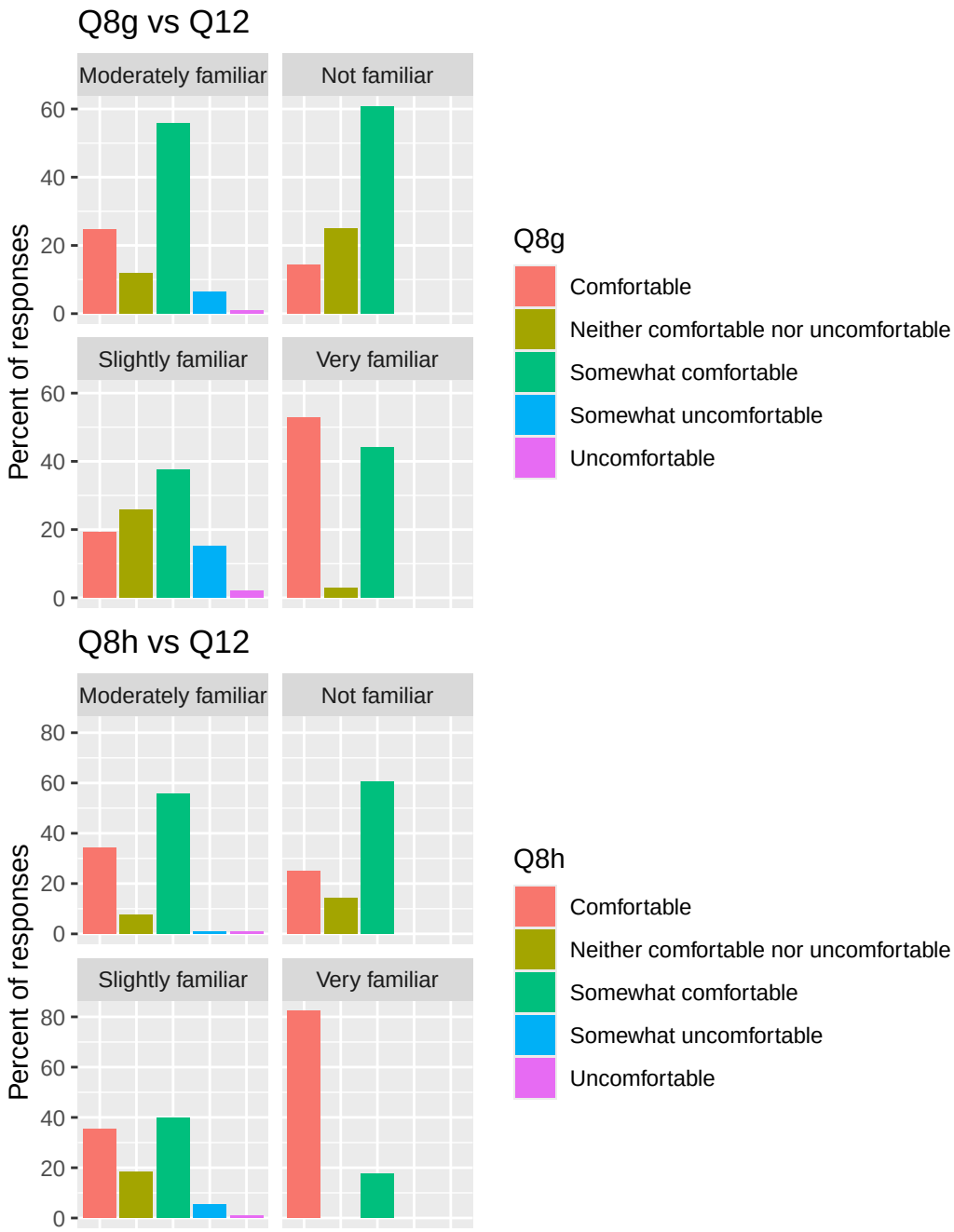
Additional Analysis Plots

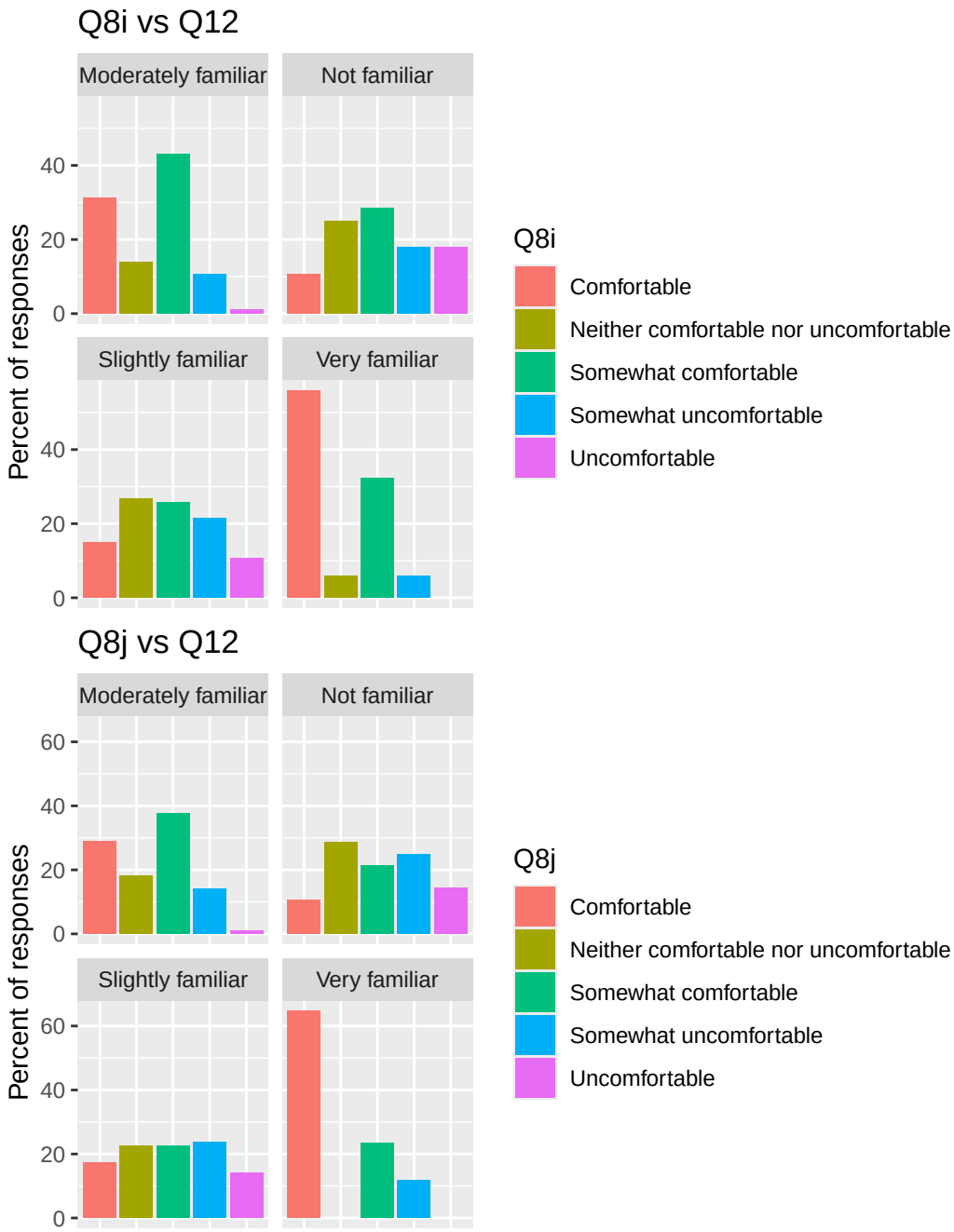
Below are plots comparing each additional analysis question.


Q8a vs Q12
Moderately familiar
Not familiar
Slightly familiar
Very familiar
Percent of responses
0
20
40
60
Q8a
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8b vs Q12
Moderately familiar
Not familiar
Slightly familiar
Very familiar
Percent of responses
0
10
20
30
40
Q8b
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable

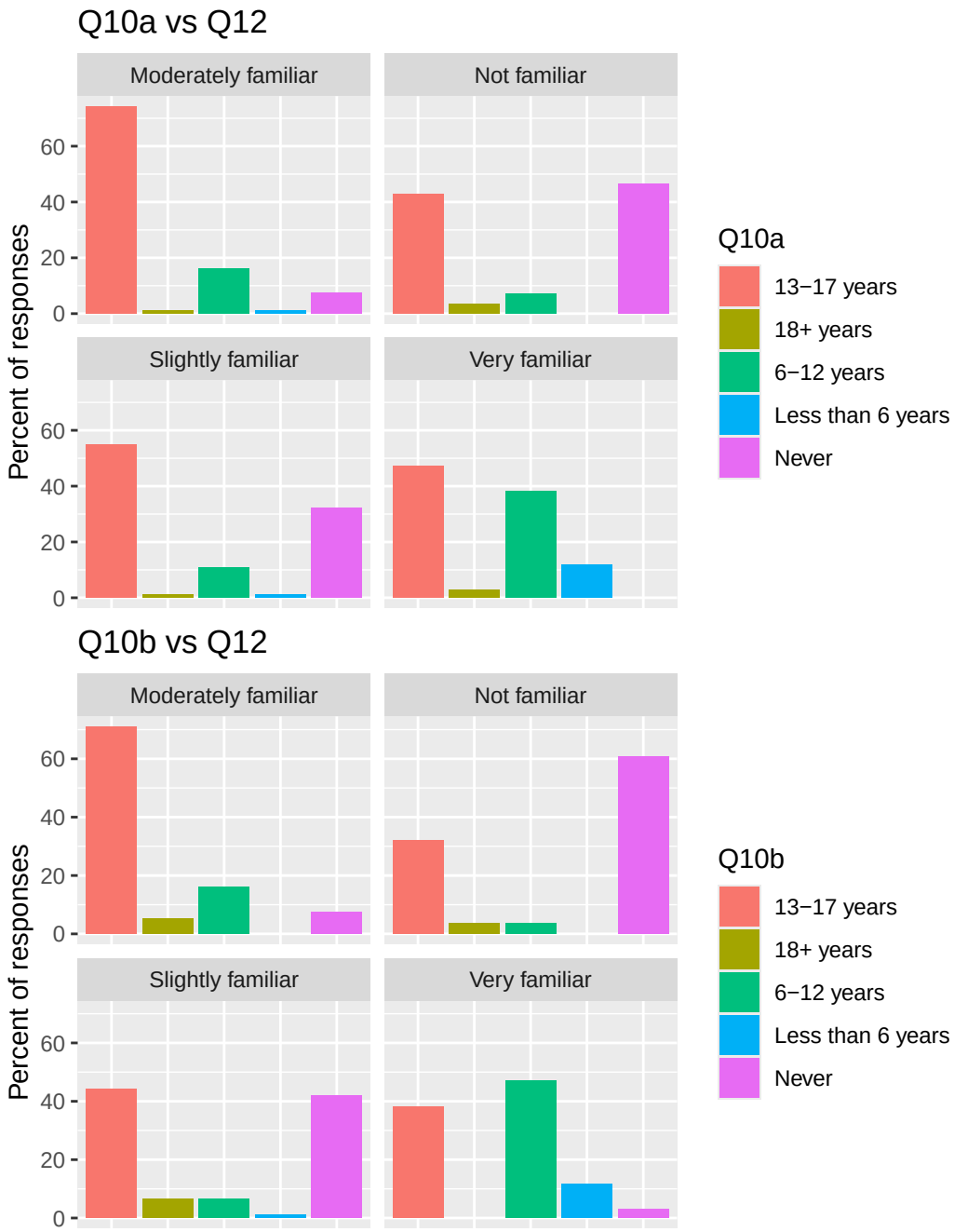

Q8c vs Q12
Moderately familiar
Not familiar
Slightly familiar
Very familiar
Percent of responses
0
10
20
30
40
Q8c
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8d vs Q12
Moderately familiar
Not familiar
Slightly familiar
Very familiar
Percent of responses
0
10
20
30
40
Q8d
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable

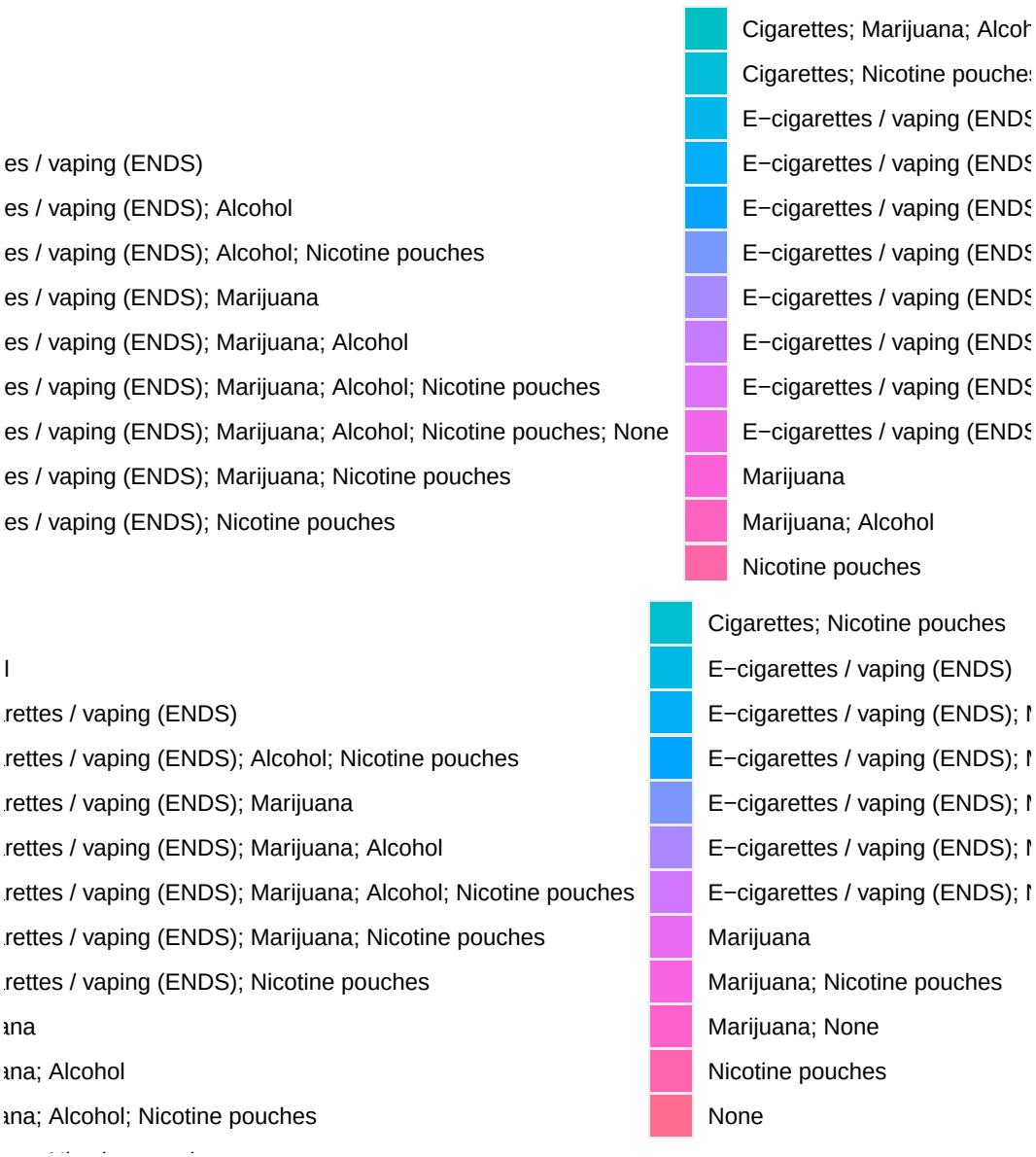

Q8e vs Q12
Moderately familiar
Not familiar
Slightly familiar
Very familiar
Percent of responses
0
20
40
60
Q8e
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8f vs Q12
Moderately familiar
Not familiar
Slightly familiar
Very familiar
Percent of responses
0
20
40
60
Q8f
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable

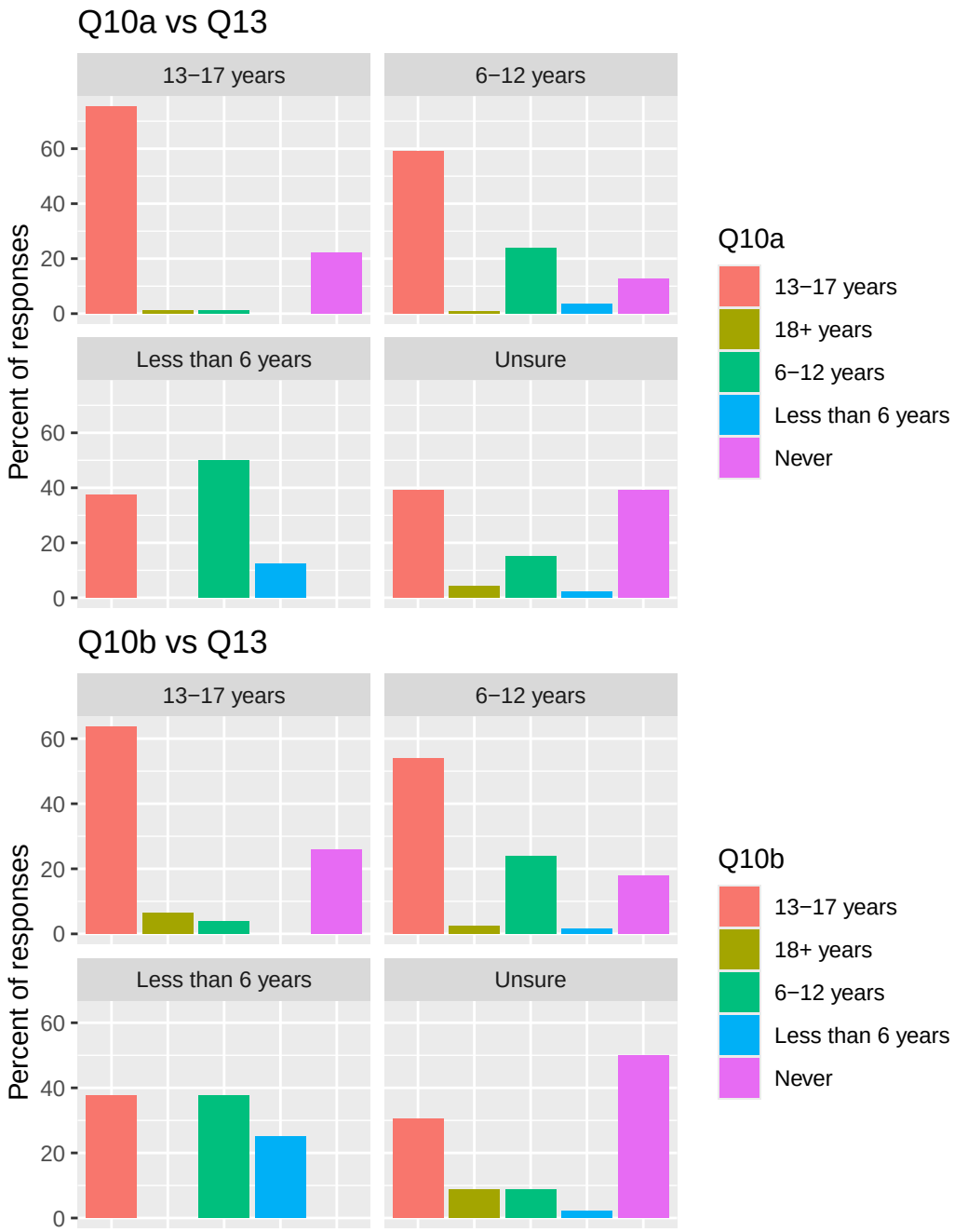

Q8g vs Q12
Moderately familiar
Not familiar
Slightly familiar
Very familiar
Percent of responses
0
20
40
60
Q8g
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8h vs Q12
Moderately familiar
Not familiar
Slightly familiar
Very familiar
Percent of responses
0
20
40
60
80
Q8h
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable


Q8i vs Q12
Moderately familiar
Not familiar
Slightly familiar
Very familiar
Percent of responses
0
20
40
Q8i
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q8j vs Q12
Moderately familiar
Not familiar
Slightly familiar
Very familiar
Percent of responses
0
20
40
60
Q8j
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable


Q8k vs Q12
Moderately familiar
Not familiar
Slightly familiar
Very familiar
Percent of responses
0
25
50
75
Q8k
Comfortable
Neither comfortable nor uncomfortable
Somewhat comfortable
Somewhat uncomfortable
Uncomfortable
Q9 vs Q12
Moderately familiar
Not familiar
Slightly familiar
Very familiar
Percent of responses
0
10
20
30
40
50
Q9
Always
Never
Often
Rarely
Sometimes


Q10a vs Q12
Moderately familiar
Not familiar
Slightly familiar
Very familiar
Percent of responses
0
20
40
60
Q10a
13−17 years
18+ years
6−12 years
Less than 6 years
Never
Q10b vs Q12
Moderately familiar
Not familiar
Slightly familiar
Very familiar
Percent of responses
0
20
40
60
Q10b
13−17 years
18+ years
6−12 years
Less than 6 years
Never


es / vaping (ENDS)
es / vaping (ENDS); Alcohol
es / vaping (ENDS); Alcohol; Nicotine pouches
es / vaping (ENDS); Marijuana
es / vaping (ENDS); Marijuana; Alcohol
es / vaping (ENDS); Marijuana; Alcohol; Nicotine pouches
es / vaping (ENDS); Marijuana; Alcohol; Nicotine pouches; None
es / vaping (ENDS); Marijuana; Nicotine pouches
es / vaping (ENDS); Nicotine pouches
Cigarettes; Marijuana; Alcoh
Cigarettes; Nicotine pouches
E−cigarettes / vaping (ENDS
E−cigarettes / vaping (ENDS
E−cigarettes / vaping (ENDS
E−cigarettes / vaping (ENDS
E−cigarettes / vaping (ENDS
E−cigarettes / vaping (ENDS
E−cigarettes / vaping (ENDS
Marijuana
Marijuana; Alcohol
Nicotine pouches
l
rettes / vaping (ENDS)
rettes / vaping (ENDS); Alcohol; Nicotine pouches
rettes / vaping (ENDS); Marijuana
rettes / vaping (ENDS); Marijuana; Alcohol
rettes / vaping (ENDS); Marijuana; Alcohol; Nicotine pouches
rettes / vaping (ENDS); Marijuana; Nicotine pouches
rettes / vaping (ENDS); Nicotine pouches
ana
ana; Alcohol
ana; Alcohol; Nicotine pouches
Cigarettes; Nicotine pouches
E−cigarettes / vaping (ENDS)
E−cigarettes / vaping (ENDS); N
E−cigarettes / vaping (ENDS); N
E−cigarettes / vaping (ENDS); N
E−cigarettes / vaping (ENDS); N
E−cigarettes / vaping (ENDS); N
Marijuana
Marijuana; Nicotine pouches
Marijuana; None
Nicotine pouches
None


Q10a vs Q13
13−17 years
6−12 years
Less than 6 years
Unsure
Percent of responses
0
20
40
60
Q10a
13−17 years
18+ years
6−12 years
Less than 6 years
Never
Q10b vs Q13
13−17 years
6−12 years
Less than 6 years
Unsure
Percent of responses
0
20
40
60
Q10b
13−17 years
18+ years
6−12 years
Less than 6 years
Never

Appendix B: Code

```
library(tidyverse)
library(zoo)
library(stringr)
library(gt)
m_headers <- read_csv("/Users/jdumanski24/Desktop/CDSC/Levine/McGannData.csv", n_max = 2, col

questions <- as.character(m_headers[1,])

answer_choices <- as.character(m_headers[2,])

questions_filled <- na.locf(ifelse(questions == "" | is.na(questions), NA, questions))

clean_headers <- paste(questions_filled, answer_choices, sep = "_")

m_rawdata <- read_csv("/Users/jdumanski24/Desktop/CDSC/Levine/McGannData.csv", skip = 2, col_

m_non_coalesced <- m_rawdata %>%
  select(c(10:ncol(m_rawdata)))

m_non_coalesced <- m_non_coalesced %>%
  mutate(
    across(
      everything(),
      ~ na_if(as.character(.), "")
    )
  )

m_questions <- m_non_coalesced %>%
  mutate(Q1 = do.call(coalesce, pick(1:6))) %>%
  mutate(Q2 = do.call(coalesce, pick(7:12))) %>%
  mutate(Q3 = do.call(coalesce, pick(13:21))) %>%
  mutate(Q4 = do.call(coalesce, pick(22:24))) %>%
  mutate(Q5 = do.call(coalesce, pick(25:27))) %>%
  mutate(Q6 = do.call(coalesce, pick(28:33))) %>%
  mutate(Q7 = do.call(coalesce, pick(34:40))) %>%
  mutate(Q8a = do.call(coalesce, pick(41:45))) %>%
  mutate(Q8b = do.call(coalesce, pick(46:50))) %>%
  mutate(Q8c = do.call(coalesce, pick(51:55))) %>%
  mutate(Q8d = do.call(coalesce, pick(56:60))) %>%
```

```
  mutate(Q8e = do.call(coalesce, pick(61:65))) %>%
  mutate(Q8f = do.call(coalesce, pick(66:70))) %>%
  mutate(Q8g = do.call(coalesce, pick(71:75))) %>%
  mutate(Q8h = do.call(coalesce, pick(76:80))) %>%
  mutate(Q8i = do.call(coalesce, pick(81:85))) %>%
  mutate(Q8j = do.call(coalesce, pick(86:90))) %>%
  mutate(Q8k = do.call(coalesce, pick(91:95))) %>%
  mutate(Q9 = do.call(coalesce, pick(96:100))) %>%
  mutate(Q10a = do.call(coalesce, pick(101:105))) %>%
  mutate(Q10b = do.call(coalesce, pick(106:110))) %>%
  #mutate(Q11a = do.call(coalesce, pick(111:116))) %>%
  #mutate(Q11b = do.call(coalesce, pick(117:122))) %>%
  mutate(Q12 = do.call(coalesce, pick(123:126))) %>%
  mutate(Q13 = do.call(coalesce, pick(127:131))) %>%
  mutate(Q14 = do.call(coalesce, pick(132:137))) %>%
  mutate(Q15 = do.call(coalesce, pick(138:141))) %>%
  mutate(Q16 = do.call(coalesce, pick(142:145))) %>%
  mutate(Q17 = do.call(coalesce, pick(146:149))) %>%
  mutate(Q18 = do.call(coalesce, pick(150:153))) %>%
  mutate(Q19 = do.call(coalesce, pick(154:157))) %>%
  mutate(Q20 = do.call(coalesce, pick(158:161)))
  #mutate(Q21 = do.call(coalesce, pick(162:169))) %>%
  #mutate(Q22 = do.call(coalesce, pick(170:175))) %>%
  #mutate(Q23 = do.call(coalesce, pick(176:183))) %>%

m_no_sata <- m_questions %>%
  select(184:204, 111:122, 205:213, 162:183)

m_coalesced <- m_no_sata %>%
  mutate(Q11a = pmap_chr(select(., 22:27), ~ paste(na.omit(c(...)), collapse = "; "))) %>%
  mutate(Q11b = pmap_chr(select(., 28:33), ~ paste(na.omit(c(...)), collapse = "; "))) %>%
  mutate(Q21 = pmap_chr(select(., 45:50), ~ paste(na.omit(c(...)), collapse = "; "))) %>%
  mutate(Q22 = pmap_chr(select(., 51:56), ~ paste(na.omit(c(...)), collapse = "; "))) %>%
  mutate(Q23 = pmap_chr(select(., 57:64), ~ paste(na.omit(c(...)), collapse = "; "))) %>%
  select(1:21, Q11a, Q11b, 34:42, Q21, Q22, Q23)

coll_ids <- m_rawdata$`Collector ID_NA`

m_coalesced <- m_coalesced %>%
  mutate(ids = coll_ids) %>%
  mutate(role = case_when(
    ids == 440127144 ~ "Dentist",
```

```
    ids == 440133230 ~ "Resident"
  ))

correct_answers <- c("Ask about tobacco or vaping use and provide counseling", "Counsel on to

m_coalesced <- m_coalesced %>%
  mutate(
    score = if_else(
      if_all(Q15:Q20, is.na),
      NA_real_,
      rowSums(
        across(Q15:Q20) == matrix(
          correct_answers,
          nrow = n(),
          ncol = length(correct_answers),
          byrow = TRUE
        ),
        na.rm = TRUE
      )
    )
  )
validation <- function(question1, question2, data){
  df <- data %>%
    transmute(
      response1 = .data[[question1]],
      response2 = .data[[question2]]
    ) %>%
    drop_na()

  comparison_res <- df %>%
    group_by(response1, response2) %>%
    summarize(counts = n(), .groups = "drop") %>%
    group_by(response2) %>%
    mutate(percent = counts / sum(counts) * 100) %>%
    ungroup()

  table_comparison <- table(df$response2, df$response1)
  #table_hours
  test <- suppressWarnings(chisq.test(table_comparison))

  if (mean(test$expected <= 5) < 0.8 | any(test$expected < 1)){
    fish_test <- fisher.test(table_comparison, simulate.p.value = TRUE)
```

```
    if (fish_test$p.value <= 0.05){
      significant <- TRUE
    }
    else {
      significant <- FALSE
    }
    return(c("question1", "question2", round(fish_test$p.value, digits = 4), significant, "fi
  }
  else{
    if (test$p.value <= 0.05){
      significant <- TRUE
    }
    else {
      significant <- FALSE
    }
  }
  return(c("question1", "question2", round(test$p.value, digits = 4), significant, "chisq"))
}
comfort_dic <- data.frame(
  q_ref = c("Q8a",  "Q8b",  "Q8c",  "Q8d",  "Q8e",  "Q8f", "Q8g",  "Q8h" , "Q8i" , "Q8j" , "(
  role = c(rep("role", 16))
)

comfort_results <- data.frame(
  Question1 = character(),
  Question2 = character(),
  P_value = numeric(),
  Significant = logical(),
  Test = character()
)

for (i in seq_len(nrow(comfort_dic))) {

  comfort_results[i, ] <- validation(
    question1 = comfort_dic$q_ref[i],
    question2 = comfort_dic$role[i],
    data = m_coalesced
  )
}

comfort_results <- comfort_results %>%
  mutate(
```

```
    Adj_P = round(p.adjust(P_value, method = "BH"), digits = 4),
    Sig_Adj = Adj_P <= 0.05,
    Question1 = comfort_dic$q_ref,
    Question2 = comfort_dic$role
  ) %>%
  arrange(Adj_P)

gt(comfort_results, caption = "Comfort/Clinical Practice")
knowledge_dic <- data.frame(
  q_ref = c("Q12", "Q13", "Q14"),
  role = c(rep("role", 3))
)

knowledge_results <- data.frame(
  Question1 = character(),
  Question2 = character(),
  P_value = numeric(),
  Significant = logical(),
  Test = character()
)

for (i in seq_len(nrow(knowledge_dic))) {

  knowledge_results[i, ] <- validation(
    question1 = knowledge_dic$q_ref[i],
    question2 = knowledge_dic$role[i],
    data = m_coalesced
  )
}

knowledge_results <- knowledge_results %>%
  mutate(
    Adj_P = round(p.adjust(P_value, method = "BH"), digits = 4),
    Sig_Adj = Adj_P <= 0.05,
    Question1 = knowledge_dic$q_ref,
    Question2 = knowledge_dic$role
  ) %>%
  arrange(Adj_P)

gt(knowledge_results, caption = "Knowledge Comparison")
quiz_dic <- data.frame(
  q_ref = c("Q15", "Q16", "Q17", "Q18", "Q19", "Q20", "score"),
```

```
  role = c(rep("role", 7))
)

quiz_results <- data.frame(
  Question1 = character(),
  Question2 = character(),
  P_value = numeric(),
  Significant = logical(),
  Test = character()
)

for (i in seq_len(nrow(quiz_dic))) {

  quiz_results[i, ] <- validation(
    question1 = quiz_dic$q_ref[i],
    question2 = quiz_dic$role[i],
    data = m_coalesced
  )
}

quiz_results <- quiz_results %>%
  mutate(
    Adj_P = round(p.adjust(P_value, method = "BH"), digits = 4),
    Sig_Adj = Adj_P <= 0.05,
    Question1 = quiz_dic$q_ref,
    Question2 = quiz_dic$role
  ) %>%
  arrange(Adj_P)

gt(quiz_results, caption = "Residents vs Dentists Difference in Quiz Answers")
env_dic <- data.frame(
  q_ref = c("Q8a",  "Q8b",  "Q8c",  "Q8d",  "Q8e",  "Q8f", "Q8g",  "Q8h" , "Q8i" , "Q8j" , "(
  Q5 = c(rep("Q5", 17))
)

env_results <- data.frame(
  Question1 = character(),
  Question2 = character(),
  P_value = numeric(),
  Significant = logical(),
  Test = character()
)
```

```r
for (i in seq_len(nrow(env_dic))) {

  env_results[i, ] <- validation(
    question1 = env_dic$q_ref[i],
    question2 = env_dic$Q5[i],
    data = m_coalesced
  )
}

env_results <- env_results %>%
  mutate(
    Adj_P = round(p.adjust(P_value, method = "BH"), digits = 4),
    Sig_Adj = Adj_P <= 0.05,
    Question1 = env_dic$q_ref,
    Question2 = env_dic$Q5
  ) %>%
  arrange(Adj_P)

gt(env_results, caption = "Environment Analysis")
dentists <- m_coalesced %>%
  filter(role == "Dentist")

prac_dic <- data.frame(
  q_ref = c("Q8a",  "Q8b",  "Q8c",  "Q8d",  "Q8e",  "Q8f", "Q8g",  "Q8h" , "Q8i" , "Q8j" , "
  Q3 = c(rep("Q3", 18))
)

prac_results <- data.frame(
  Question1 = character(),
  Question2 = character(),
  P_value = numeric(),
  Significant = logical(),
  Test = character()
)

for (i in seq_len(nrow(prac_dic))) {

  prac_results[i, ] <- validation(
    question1 = prac_dic$q_ref[i],
    question2 = prac_dic$Q3[i],
    data = dentists
  )
```

```
}

prac_results <- prac_results %>%
  mutate(
    Adj_P = round(p.adjust(P_value, method = "BH"), digits = 4),
    Sig_Adj = Adj_P <= 0.05,
    Question1 = prac_dic$q_ref,
    Question2 = prac_dic$Q3
  ) %>%
  arrange(Adj_P)

gt(prac_results, caption = "Dental Practice Analysis")


addt_dic <- data.frame(
  q_ref = c("Q8a",  "Q8b",  "Q8c",  "Q8d",  "Q8e",  "Q8f", "Q8g",  "Q8h" , "Q8i" , "Q8j" , "(
  q_group = c(rep("Q12", 14), "Q9", "Q9", "Q13", "Q13")
)

addt_results <- data.frame(
  Question1 = character(),
  Question2 = character(),
  P_value = numeric(),
  Significant = logical(),
  Test = character()
)

for (i in seq_len(nrow(addt_dic))) {

  addt_results[i, ] <- validation(
    question1 = addt_dic$q_ref[i],
    question2 = addt_dic$q_group[i],
    data = m_coalesced
  )
}

addt_results <- addt_results %>%
  mutate(
    Adj_P = round(p.adjust(P_value, method = "BH"), digits = 4),
    Sig_Adj = Adj_P <= 0.05,
    Question1 = addt_dic$q_ref,
    Question2 = addt_dic$q_group
```

```
  ) %>%
  arrange(Adj_P)

gt(addt_results, caption = "Additional Analysis")
stack_bar <- function(q_reference, q_group, data){
  bar_df <- data %>%
    group_by(.data[[q_reference]], .data[[q_group]]) %>%
    summarize(counts = n(), .groups = "drop") %>%
    drop_na() %>%
    group_by(.data[[q_group]]) %>%
    mutate(percent = counts / sum(counts) * 100) %>%
    ungroup()

  p <- ggplot(bar_df, aes(x = .data[[q_reference]], y = percent, fill = .data[[q_reference]]
    geom_col() +
    facet_wrap(vars(.data[[q_group]])) +
    labs(
      title = paste(q_reference, "vs", q_group, sep = " "),
      x = NULL,
      y = "Percent of responses"
    ) +
    theme(
      axis.text.x = element_blank(),
      axis.ticks.x = element_blank()
    )
  return(p)
}
plots_comfort <- pmap(comfort_dic,
                      function(q_ref, role){
                        stack_bar(
                          q_reference = q_ref,
                          q_group = role,
                          data = m_coalesced
                        )
                      })
walk(plots_comfort, print)
plots_knowledge <- pmap(knowledge_dic,
                        function(q_ref, role){
                          stack_bar(
                            q_reference = q_ref,
                            q_group = role,
                            data = m_coalesced
```

```
                        )
                      })
walk(plots_knowledge, print)
plots_quiz <- pmap(quiz_dic,
                      function(q_ref, role){
                        stack_bar(
                          q_reference = q_ref,
                          q_group = role,
                          data = m_coalesced
                        )
                      })
walk(plots_quiz, print)
plots_env <- pmap(env_dic,
                      function(q_ref, Q5){
                        stack_bar(
                          q_reference = q_ref,
                          q_group = Q5,
                          data = m_coalesced
                        )
                      })
walk(plots_env, print)
plots_dentists <- pmap(prac_dic,
                      function(q_ref, Q3){
                        stack_bar(
                          q_reference = q_ref,
                          q_group = Q3,
                          data = dentists
                        )
                      })
walk(plots_dentists, print)
plots_addt <- pmap(addt_dic,
                      function(q_ref, q_group){
                        stack_bar(
                          q_reference = q_ref,
                          q_group = q_group,
                          data = m_coalesced
                        )
                      })
walk(plots_addt, print)
```